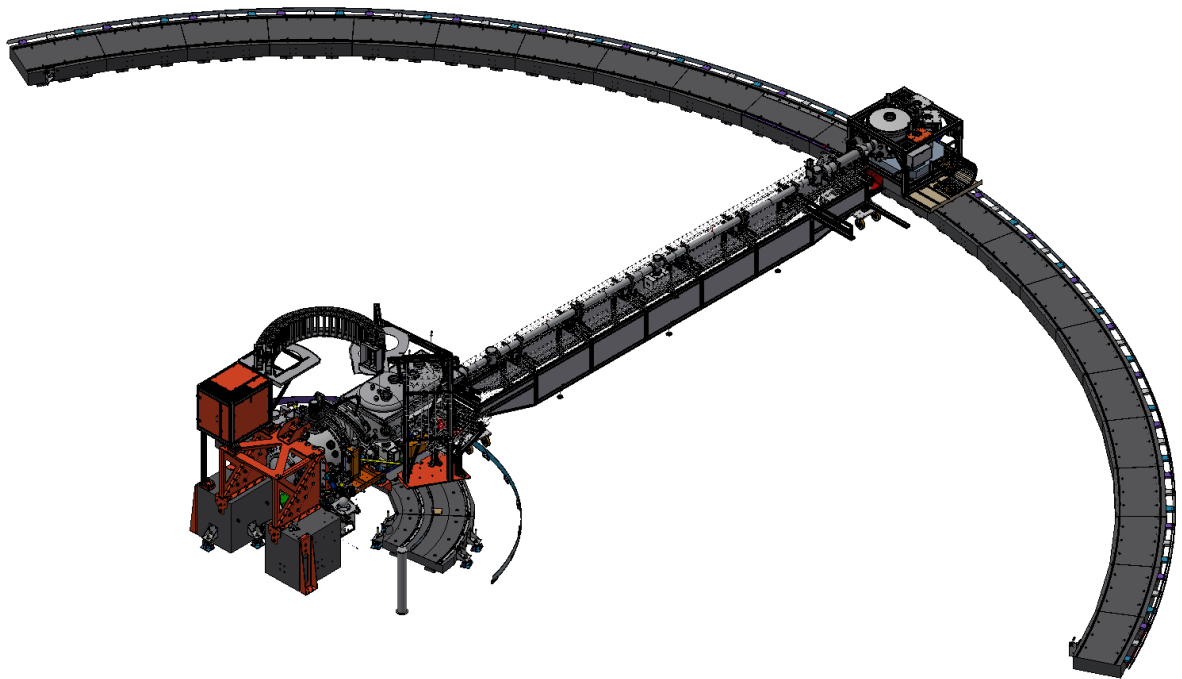


On-Site Acceptance Tests Protocol

Spectrometer Arm System for SIX Beamline
at the National Synchrotron Light Source II
at Brookhaven National Laboratory



BESTEC Preliminary OSAT Protocol - Project P442

Version 1.3

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List of Abbreviations

UHV	Ultra High Vacuum
SAS	Spectrometer Arm System
EPS	Equipment Protection System
SMR	Spherically Mounted Retroreflector

SC - sample chamber

MHT1	Manipulator Horizontal Translation 1
MHT2	Manipulator Horizontal Translation 2
MVT	Manipulator Vertical Translation
MR	Manipulator Rotation

M5 mirror system

M5R2T	M5 Mirror 2-Theta Rotation
M5T1->M5T6	M5 Mirror Tripod Axis 1 ->6
PTX	Pivot Translation X
PTY	Pivot Translation Y
M5M	M5 Mirror Mask
TC1	Thermocouple at Mirror Holder
TC2	Thermocouple at Mirror Holder
TC3	Thermocouple at Rotating Support
TC4	Thermocouple at Rotating Support
TC5	Thermocouple at Support Pillar
TC6	Thermocouple at Support Pillar
CSW-M5T1	Collision Switch at M5 Holder
PD1	Photodiode
PD2	Photodiode
PD3	Retractable Photodiode

3AA - 3 m arm and optics tank assembly

3AA	3m Arm and Optics Tank Assembly
M6P	M6 Mirror Rotation (pitch)
M6TZ	M6 Mirror Translation
M6M	M6 Mirror Mask
M7P	M7 Mirror Rotation (pitch)
GP	Grating Rotation (pitch)

GT	Grating Translation
BTZ	Optics Base Plate Translation
M7M	M7 Mirror Mask
3AAR2T	3AA 2-Theta Rotation
OVY	Wedge Leveler Outboard
IVY	Wedge Leveler Inboard
TC7	Thermocouple at Bearing Block Left
TC8	Thermocouple at Bearing Block Right
TC9	Thermocouple at Gratings
TC10	Thermocouple at M7 Grating Support
TC11	Thermocouple at M6 Grating Support
TC12	Thermocouple at 2Theta Traveler
CSW-GP	Collision Switch Grating Rotation (Pitch)
FSW-GT	Free Switch Grating Translation
CSW12-3AAR2T	Collision Switch
CSW34-3AAR2T	Collision Switch
FSW-3AAR2T	Free Switch

DAA - detector arm and stand assembly

DAA	Detector Arm and Stand Assembly
DAAHUL	Detector Arm Height Upstream Left
DAAHUR	Detector Arm Height Upstream Right
DAAHD	Detector Arm Height Downstream
DAAR2T	DAA 2-Theta Rotation
DT	Detector Translation
CSW-DAAR2T	Collision Switch
AT	Aperture Top
AB	Aperture Bottom
AL	Aperture Left
AR	Aperture Right
FWD KSW	Forward Kill Switch
REV KSW	Reverse Kill Switch
FWD LSW	Forward Limit Switch
REV LSW	Reverse Limit Switch

BESTEC GmbH holds the rights for the design of the system. All given technical information shall be handled careful and secure.

1 Coordinate system and conventions

This document uses a global Cartesian coordinate system to specify various orientations and directions. The origin of this coordinate system, which is illustrated in Figure 1, is the following:

- The height (Y) is set as the height of the electron beam in the storage ring.
- It is located in the horizontal plane (X, Z) at the ideal location of the sample θ axis of rotation, the 3-m arm 2θ axis of rotation, and the detector arm 2θ axis of rotation, the three of which ideally coincide.

The positive Z axis runs parallel to the inboard wall of the experimental area in the nominal direction of beam propagation. The positive X axis points outboard and is in the horizontal plane. The positive Y axis points vertically upward. 2θ angles are measured positive as counterclockwise from the positive Z axis when looking from the top.

One set of rotation axes is defined as rotation about each of the Cartesian axes. The designation is (ax, ay, az), meaning “angle about x,” etc. The sign of these rotations is right-handed, except for the angle ϕ , which is taken as positive in the upwards direction, with the horizontal taken as the zero reference.

A second set of rotation axes are the named rotations (e.g., pitch, theta, etc.), depending on the requirements of the component. The relationships between these rotations and ax, ay, and az are listed in each of the tables of motions, where applicable. Note that the detector local coordinate system ($X_{det}Y_{det}Z_{det}$) rotates with 2θ .

The optical components along the spectrometer arm system have individual local coordinate systems, as illustrated in Figure 1-5 for M6, M7 and the grating and Figure 1-6 for M5. The origin of the M5, M6, M7 and grating coordinate systems is set at the pole of the respective optic.

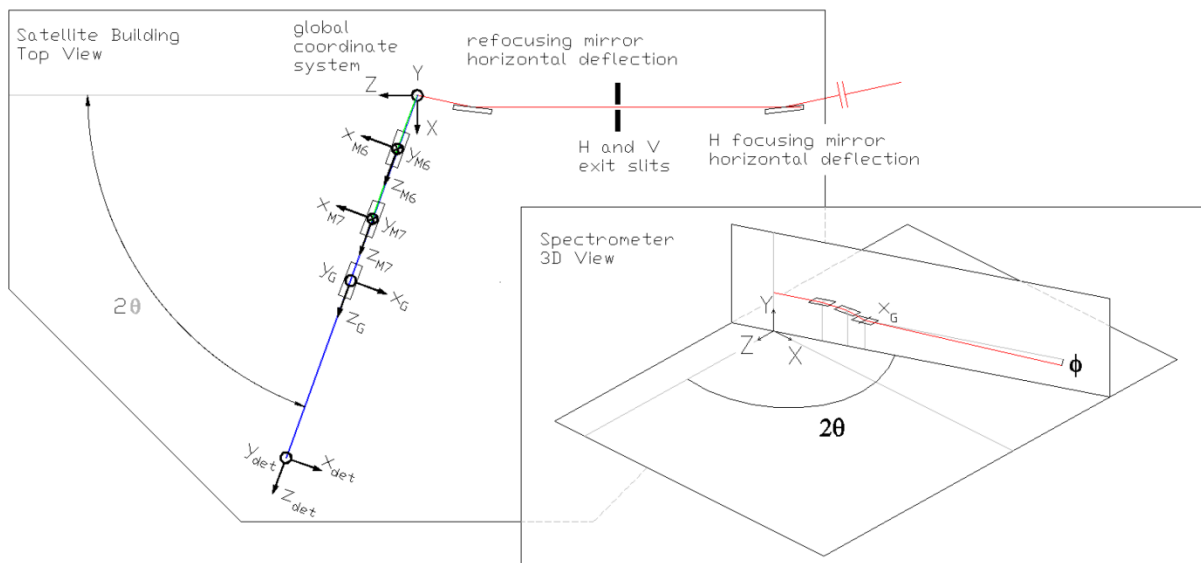


Figure 1: Coordinate system sketch.

The global coordinate system is referred to as XYZ, the M6 coordinate system as $x_{M6}y_{M6}z_{M6}$, the grating coordinate system as $x_Gy_Gz_G$ and the detector coordinate system (which rotates with 2θ and stays at $\phi = -1$) as $x_{det}y_{det}z_{det}$. The M6 and the grating coordinate systems are understood to be oriented with the optical face of the element. The angle the detector makes in a vertical plane about the center of pitch rotation of the gratings (x_G) is denoted by ϕ and is taken as positive upwards with the zero at horizontal.

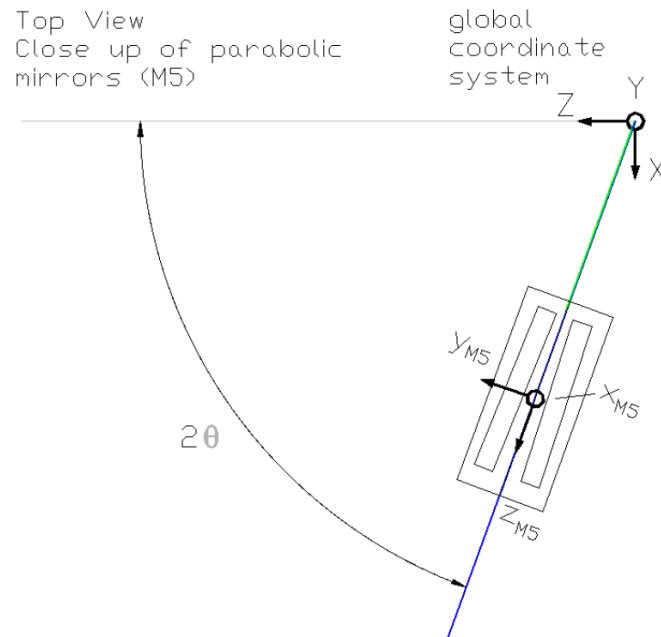


Figure 2: Sketch of the two M5 mirrors on their common baseplate and their coordinate system. $x_{M5}y_{M5}z_{M5}$ are defined with respect to the baseplate of the M5 mirrors.

2 Commissioning of the system at Bestec - FAT

2.1 M5 mirror system

2.1.1 Test procedure

The parameter values listed throughout this document are guaranteed when the room temperature is stabilized to a constant value between 20 °C and 25 °C with ± 0.5 K accuracy and when the vibration level below 200 Hz does not exceed 2×10^{-5} g /1% bandwidth (measured with an accelerometer at the floor on-site).

Metrology and alignment at Bestec GmbH were carried out using Bestec motion control system (Phytron ZMX+ motor drivers and National Instruments NI7344 motion controllers) and motion control software together with the diagnostic elements included in the systems and Bestec diagnostics tools (Autocollimator Möller Wedel - Elcomat direct 500T/65, Theodolite Leica TM6100A, Mass spectrometer SRS RGA200).

Since some of the parameter values are at the precision limit of the diagnostic tools and setups and subject to statistical errors, all tests are carried out in a way suitable for statistical analysis and all values given for repeatability and stability are standard deviations (1σ).

Parameter values, which can be guaranteed using the Bestec control system (in Closed-Loop-Control) are marked with "CLC". The system parameters will be determined and optimized using the procedures described below.

Mirror rotation and translation ranges

- Limit and kill switches for all axes are tested and set.
- Rotation and translation ranges are determined.

Positioning and stability scans

- Mirror rotation angle scans optimized for high resolution and stability are performed (encoder signal is recorded as a function of time).

Repeatability of mirror pitch and roll upon rotation

- An auxiliary mirror is temporary fixed at the tested axis.
- The optical element (mirror dummy) is moved several times and positioned each time to the same encoder position.
- The rotation repeatability is tested with an additional angle measurement system (Autocollimator Möller Wedel, Elcomat direct 500T/65) which reads the orientation after each new positioning.

Circle of confusion and wobble of M5 2Theta

- SoC: A precision sphere is fixed temporarily in the M5 rotation center and a precision digital dial gauge is fixed at the M5 rotation platform touching the precision sphere.
- Wobble: A precision water level (10 μ rad resolution) is set on the M5 2Theta rotation platform and read at different 2Theta angles.

2.1.2 Technical specifications

Table 1: Overview of technical specifications (M5)

General	
Mirror unit	Mirror mechanics (tripod) included in the sample chamber
Beam height	1419.25 mm
Number of mirrors	2 collimating mirrors
Substrate dimensions (L x W x H) Active optical surface (L x W)	260 mm x 50 mm x 72.5 mm 250 mm x 10 mm
Grazing angle	88.5°
M5 mirror rotation 2-theta (M5R2T)	
Motor	Phytron ZSS57.200.2,5-GPL52/50
Step resolution (full step)	3.6×10^{-3} °/step
Encoder	Renishaw Resolute Encoder scale: RESA30USA350B Readhead: RA26BVA350B15V
Encoder resolution	5.36441×10^{-6} °/count
Limit switches	Huber LS01
Circle of minimum confusion (M5 2Theta rotation stage)	Max. radius = 2.99 μm RMS radius = 2.14 μm
Tangential wobble of the system	~8.6 μrad
M5 mirror pitch	
Repeatability of mirror rotation upon pitch variation (CLC)	Pitch: 0.42 μrad Roll: 0.2 μrad
Actuator drives for all six tripod axes (M5T1 -> M5T6)	
Travel range	26 mm
Long-term (>25 minutes) tripod axis stability	0.0065 μm
Motor	PI PiezoMike N-470.42U
Typical step size	20 nm (max. 30 nm)
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B50V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
Pivot translation X (PTX)	

Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Pivot translation Z (PTZ)	
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
M5 masks (M5M)	
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1 mm
Step resolution (full step)	5 μm /step
Limit switches	2 x Huber LS01
Retractable photodiode (PD3)	
Motor	PI PiezoMike N-470.42U
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Limit switches	2 x Huber LS01
Circle of minimum confusion (M5 2Theta rotation stage)	Max. radius = 2.99 μm RMS radius = 2.14 μm
Tangential wobble of the system	~8.6 μrad

2.1.3 Test results

Repeatability of the mirror M5 rotation (pitch, roll) upon mirror pitch variation

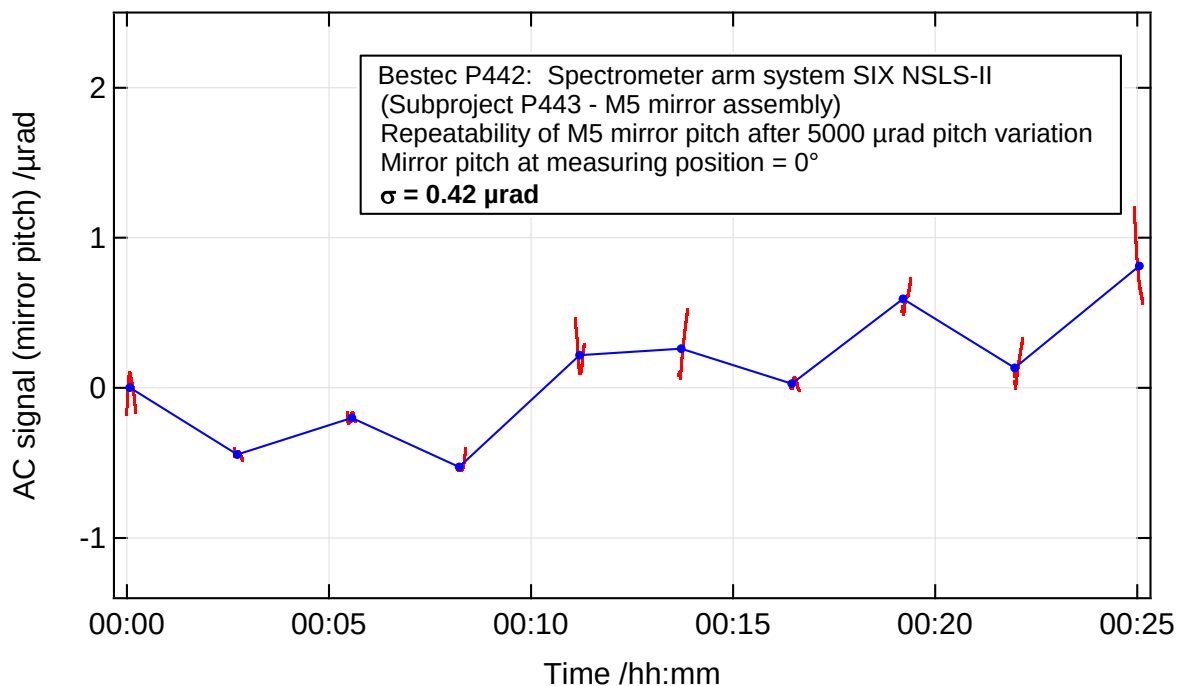


Figure 3: Autocollimator raw data and processed data showing the repeatability of M5 mirror pitch upon mirror pitch variation (0° measuring position).

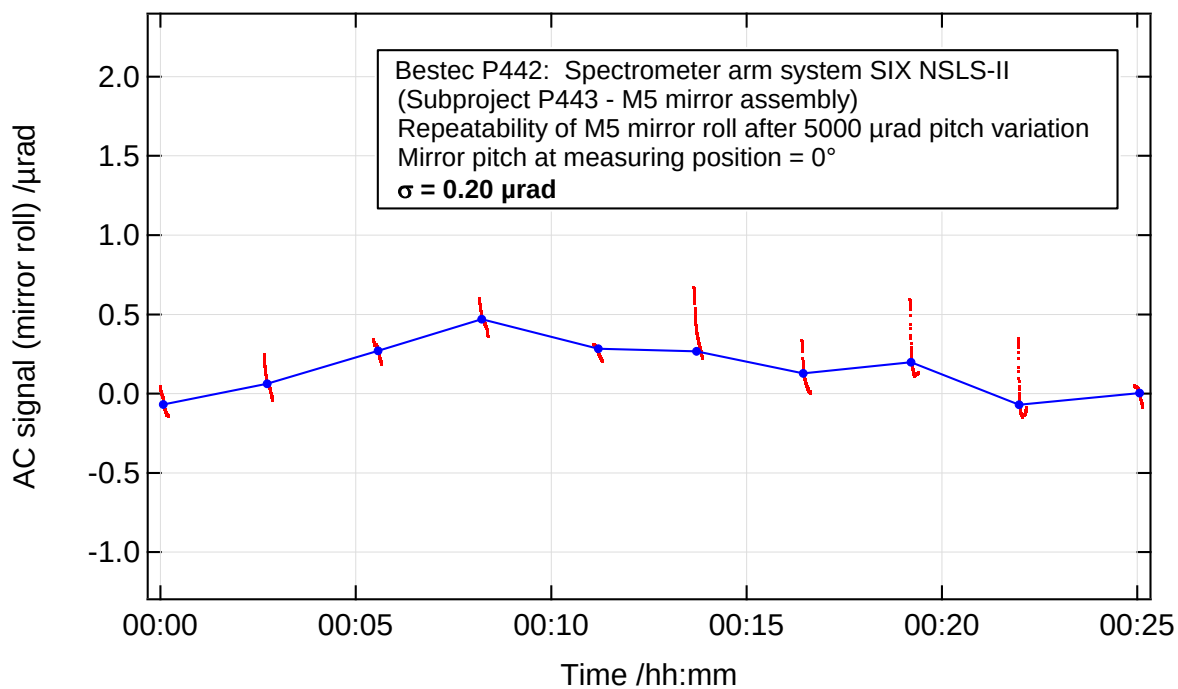


Figure 4: Autocollimator raw data and processed data showing the repeatability of M5 mirror roll upon mirror pitch variation (0° measuring position).

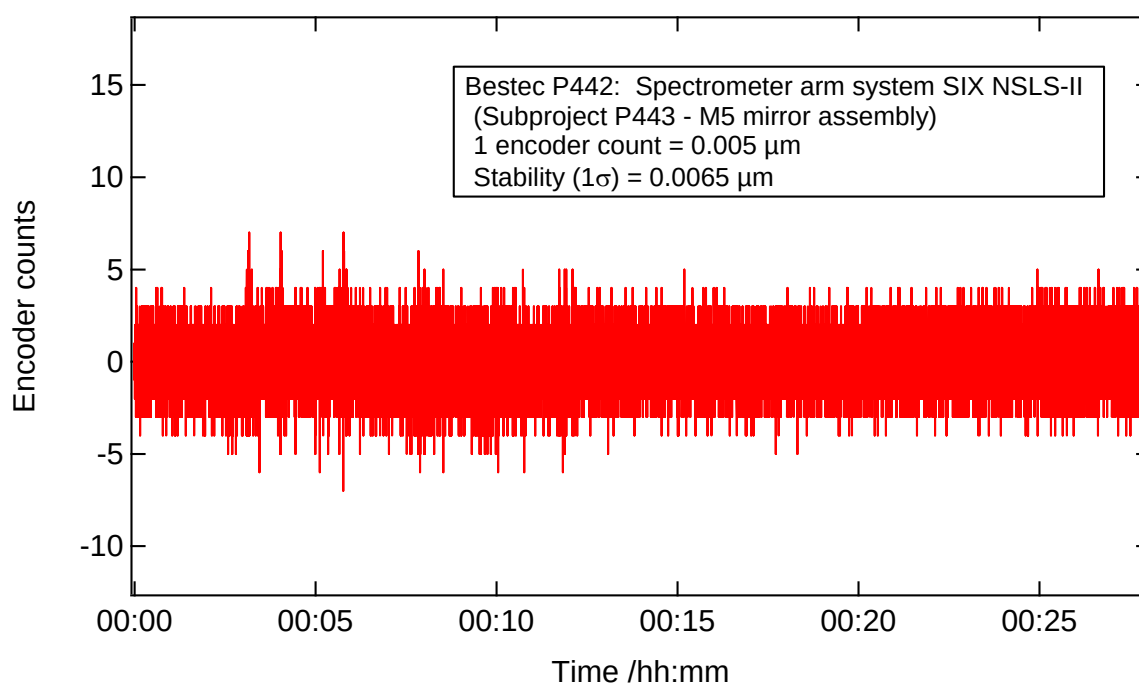
Long-term stability of tripod axis

Figure 5: Long-term stability of a representative tripod axis. Encoder data are neither averaged nor filtered.

Circle of minimum confusion of M5 2Theta

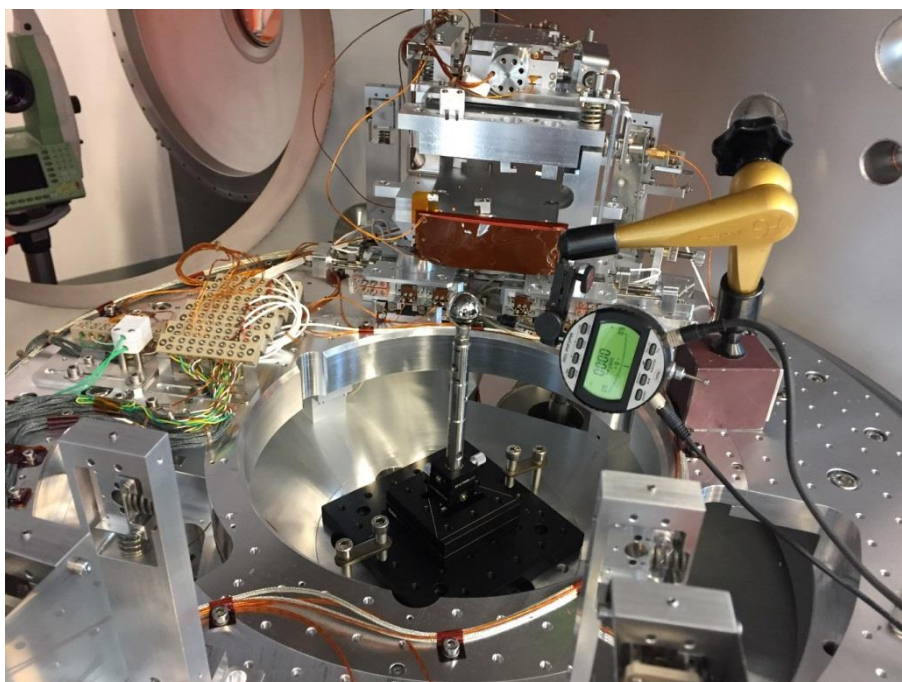


Figure 6: Test setup for the determination of the circle of minimum confusion of the M5 2Theta rotation stage.

M5 2Theta Circle of minimum confusion

Max. Radius = 2.99 μm

RMS Radius = 2.14 μm

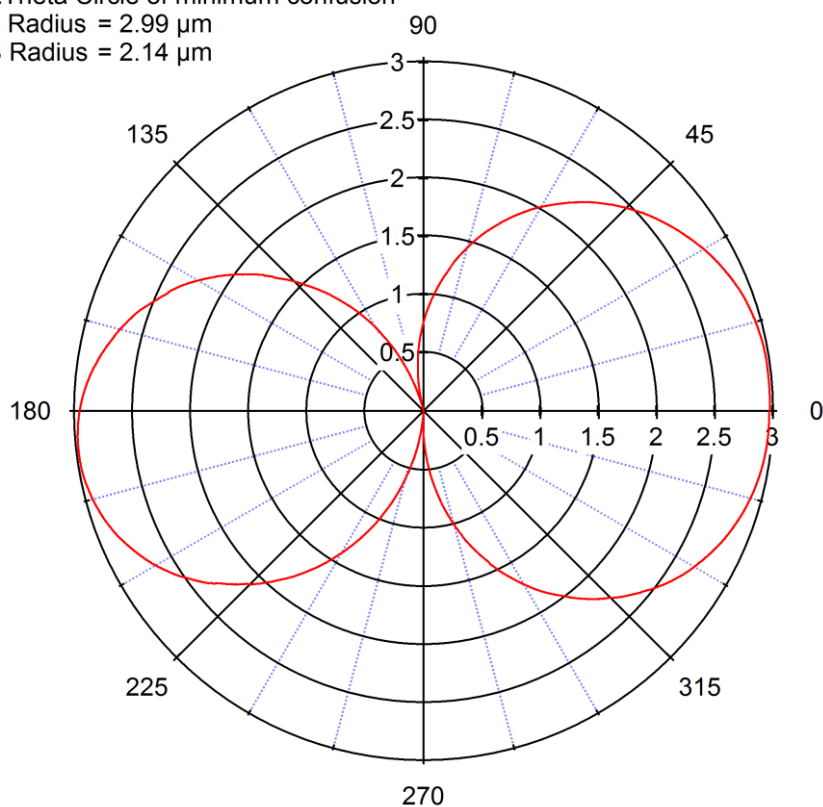


Figure 7: Circle of minimum confusion of the M5 2Theta rotation stage.

Wobble of M5 2Theta

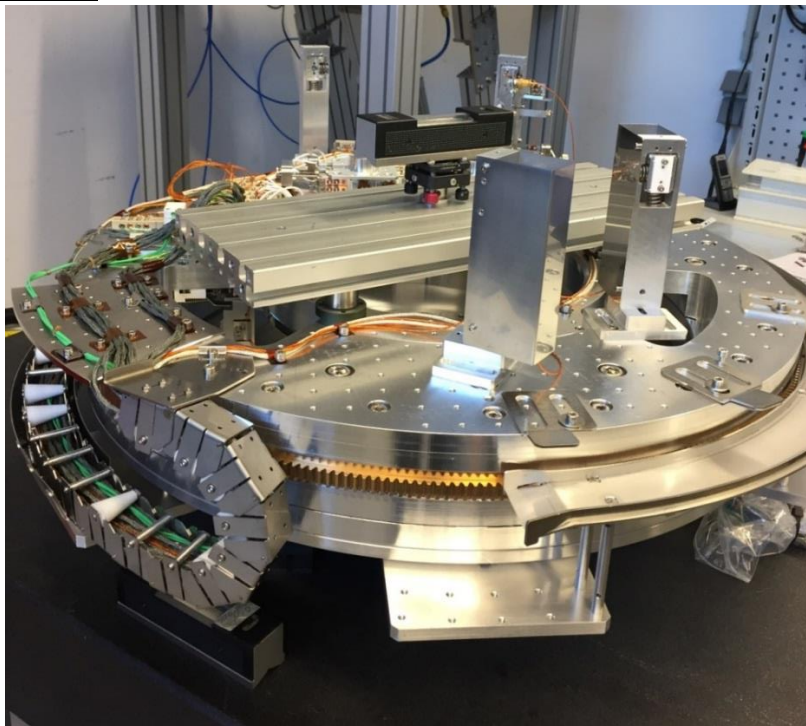


Figure 8: Test setup for the determination of the tangential wobble of the M5 2Theta rotation stage.

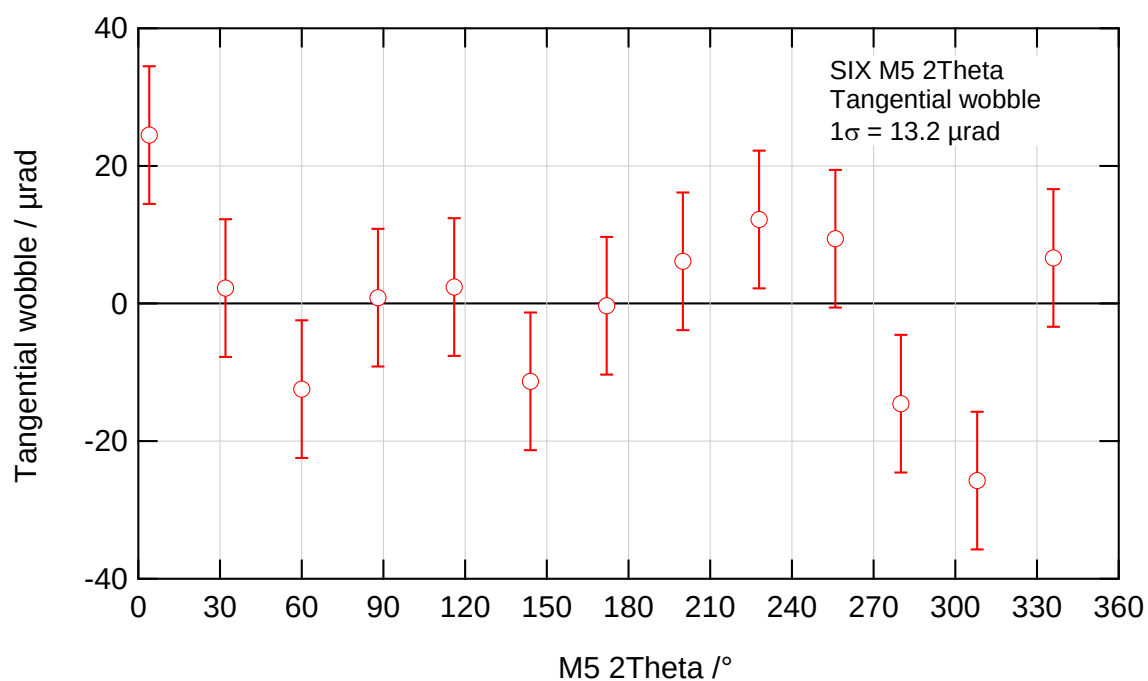


Figure 9: Tangential wobble of the M5 2Theta rotation stage.

Taking into account the uncertainty of the water level reading of $\sim 10 \mu\text{rad}$ the tangential wobble of the system only is $\sim 8.6 \mu\text{rad}$.

2.2 3AA - 3m arm and optics tank assembly

2.2.1 Test procedure

The parameter values listed throughout this document are guaranteed when the room temperature is stabilized to a constant value between 20 °C and 25 °C with ± 0.5 K accuracy and when the vibration level below 200 Hz does not exceed 2×10^{-5} g /1% bandwidth (measured with an accelerometer at the floor on-site).

Metrology and alignment at Bestec GmbH were carried out using Bestec motion control system (Phytron ZMX+ motor drivers and National Instruments NI7344 motion controllers) and motion control software together with the diagnostic elements included in the systems and Bestec diagnostics tools (Autocollimator Möller Wedel - Elcomat direct 500T/65, Theodolite Leica TM6100A, Mass spectrometer SRS RGA100).

Since some of the parameter values are at the precision limit of the diagnostic tools and setups and subject to statistical errors, all tests are carried out in a way suitable for statistical analysis and all values given for repeatability and stability are standard deviations (1σ).

Parameter values, which can be guaranteed using the Bestec control system (in Closed-Loop-Control) are marked with "CLC". The PGM parameters will be determined and optimized using the procedures described below.

Mirror (M6, M7) rotation and translation ranges

- Horizontal position of the grating and mirror are determined with the water level.
- Limit and kill switches for all axes are tested and set.
- Rotation and translation ranges are determined.

Gratings rotation and translation ranges

- Limit and kill switches for all axes are tested and set.
- Rotation and translation ranges are determined.

Positioning and stability scans (M6, M7 and gratings)

- Mirrors/gratings rotation angle scans optimized for high resolution and stability are performed (encoder signal is recorded as a function of time).

Repeatability of mirror (M6, M7) and gratings pitch and roll upon pitch variation

- An auxiliary mirror is temporary fixed at the tested axes.
- The optical elements (dummies) are moved several times and positioned each time to the same encoder position.
- The rotation repeatability is tested with an additional angle measurement system (Autocollimator Möller Wedel, Elcomat direct 500T/65), which reads the orientation after each new positioning.

2.2.2 Technical specifications

The M6 and M7 mirrors with their holders and the three gratings and their holders are housed together into one UHV chamber. Table 2 lists all technical parameters of the M6 mirror. Table 3 lists all technical parameters of the M7 mirror, of the gratings and of the vacuum vessel.

Table 2: Overview of technical specifications (M6)

M6 mirror rotation (M6P)	
Range relative to horizontal	-58715.3 μ rad to 23873.7 μ rad (-3.36° to 1.4°)
Offset	327890 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.13 μ rad Roll: 0.07 μ rad
M6 mirror pitch stability (CLC)	0.016 μ rad
Linear drive	PE2.4.0100GRT12x1
Motor	Phytron ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	240
Sine-bar connecting rod	115 mm
Angle offset	35°
Step resolution (full step)	0.42 μ rad /step
Encoder	Renishaw Tonic Encoder scale: RGSZ20-S Readhead (2x): T2641-30M Interpolator (2x): TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Reverse limit switch	-5719787 counts
Forward limit switch	2786874 counts
M6 mirror translation (M6TZ)	
Offset	475859 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.08 μ rad Roll: 0.16 μ rad
M7 mirror pitch stability (CLC)	0.022 μ rad
Motor	PI PiezoMike N-470.42U

Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
M6 masks (M6M)	
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1.25 mm
Step resolution (full step)	6.25 μm /step
Limit switches	2 x Huber LS01

Table 3: Overview of technical specifications (vacuum vessel, M7, gratings)

General	
Included components	M6 mirror mechanics M7 mirror mechanics Gratings mechanics M7 masks
Total weight of the system	~2000 kg (without lower granite block)
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	Stadium shaped
Main flange (chamber lid)	At the top (wire sealing)
Number of deflecting mirrors	1
Deflecting direction	Downward in vertical plane
Substrate dimensions (L x W x H) Active optical surface (L x W)	540 mm x 60 mm x 70 mm 525 mm x 40 mm
Number of gratings	3
Substrate dimensions (L x W x H) Active optical surface (L x W)	220 mm x 55 mm x 50 mm 200 mm x 40 mm
Base pressure of empty chamber after bakeout	3.5×10^{-10} mbar
M7 mirror and grating axis position (Reimer-Torge parameters)	
Beam offset	20 mm

Horizontal position of the plane mirror axis with respect to the grating axis	0.53 mm (closer to the source)
Vertical position of the plane mirror axis with respect to the grating axis	10.072 mm (below grating rotation axis)
M7 mirror rotation (M7P)	
Range relative to horizontal	-211687.7375 μ rad to 21537.6247 μ rad (-12.12° to 1.23°)
Offset	475859 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.08 μ rad Roll: 0.16 μ rad
M7 mirror pitch stability (CLC)	0.022 μ rad
Linear drive	ERO MOLS 100.100 (with threaded spindle 12 x 1)
Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	495 mm
Sine-bar connecting rod	115 mm
Angle	15°
Step resolution (full step)	0.20 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206 Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Reverse limit switch	-21328063 counts
Forward limit switch	2694243 counts
Gratings rotation (GP)	
Range relative to horizontal	-231376.3 μ rad to 20339.5 μ rad (-13.25° to 1.16°)
Offset	-1081502 counts
Repeatability of grating rotation upon rotation (CLS)	Pitch: 0.10 μ rad Roll: 0.12 μ rad
Repeatability of grating rotation	Pitch: 0.08 μ rad Roll: 0.09 μ rad

upon translation (CLS)	
Grating pitch stability	0.026 μ rad
Linear drive	PE2.4.0100GRT12x1
Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	340 mm
Sine-bar connecting rod	115 mm
Angle offset	12°
Step resolution	0.3 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206 Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Collision switches	2 x Huber LS01
Reverse limit switch	-24913358 counts
Forward limit switch	1013475 counts
Gratings translation (GT)	
Range	0 mm to 173.45 mm
Linear drive	Bestec linear drive
Motor	Phytron ZSS57.200.2,5
Spindle pitch	2 mm
Step resolution (full step) Belt gear 2:1	10 μ m /step 5 μ m /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μ m /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Reverse limit switch	259896128 counts

Forward limit switch	294587289 counts
Optics base plate translation (BTZ)	
Range	
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
2-theta rotation (3AAR2T)	
Motor	ZSS57.200.2,5-GPL52/144
Step resolution (full step)	7.78137×10^{-4} °/step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B10A
Encoder resolution	4.98311×10^{-6} °/count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Wedge leveler outboard (OVY)	
Range	-5.09 mm to 5.07 mm
Offset	19736085 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Reverse limit switch	19634140 counts

Forward limit switch	19837650 counts
Wedge leveler inboard (IVY)	
Range	-5.05 mm to 5.04 mm
Offset	17718638 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Reverse limit switch	17617467 counts
Forward limit switch	17819516 counts
M7 masks (M7M)	
Motor	Phytron ZSS57.200.2,5
Linear drive	VAb LD40-100
Spindle pitch	3 mm
Step resolution (full step)	15 μm /step
Limit switches	2 x Schaltbau 880

2.2.3 Test results

Repeatability of the mirror M6 rotation (pitch, roll) upon mirror pitch variation

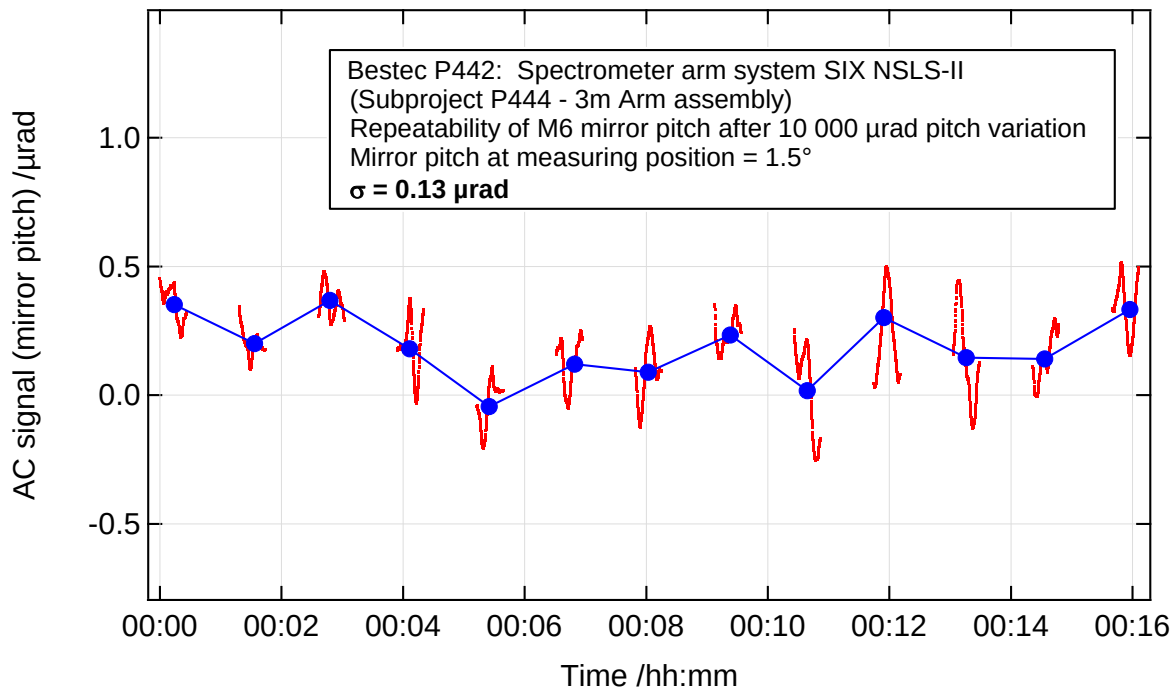


Figure 10: Autocollimator raw data and processed data showing the repeatability of M6 mirror pitch upon mirror pitch variation (1.5° measuring position).

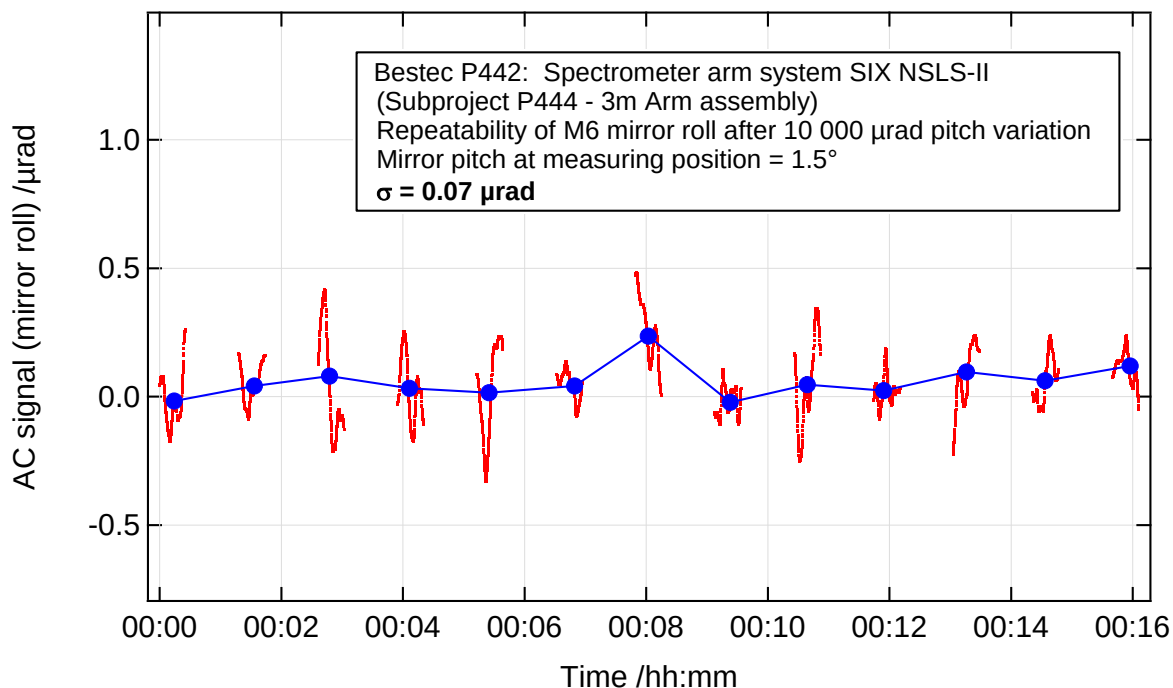


Figure 11: Autocollimator raw data and processed data showing the repeatability of M6 mirror roll upon mirror pitch variation (1.5° measuring position).

Repeatability of the mirror M7 rotation (pitch, roll) upon mirror pitch variation

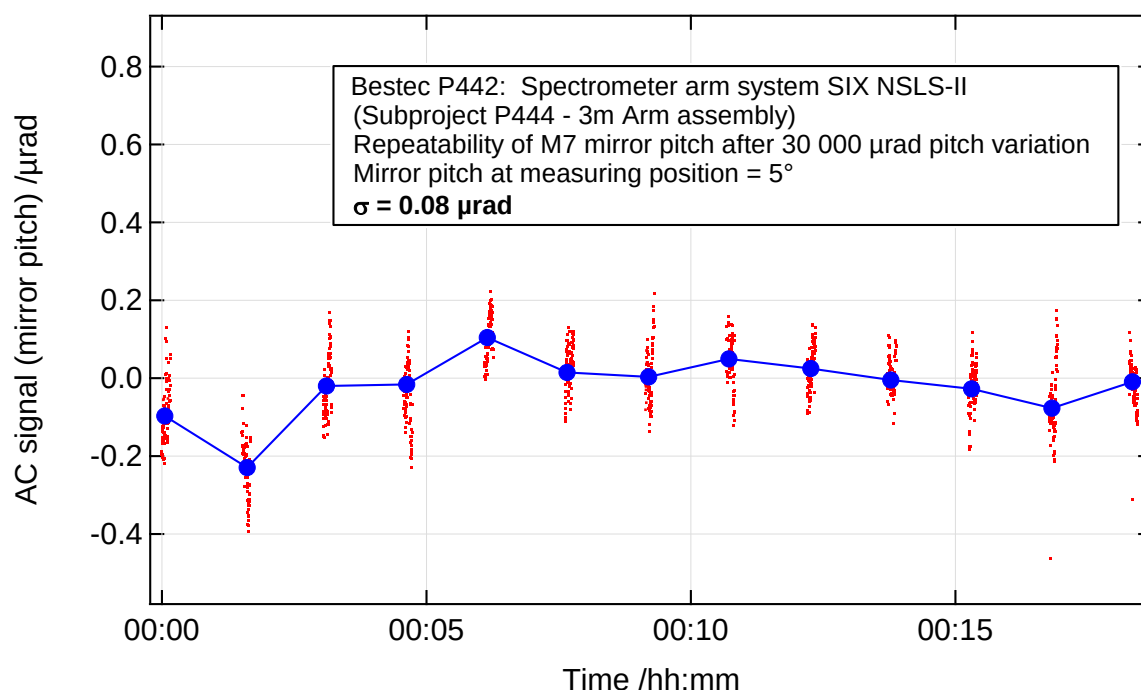


Figure 12: Autocollimator raw data and processed data showing the repeatability of M7 mirror pitch upon mirror pitch variation (5° measuring position).

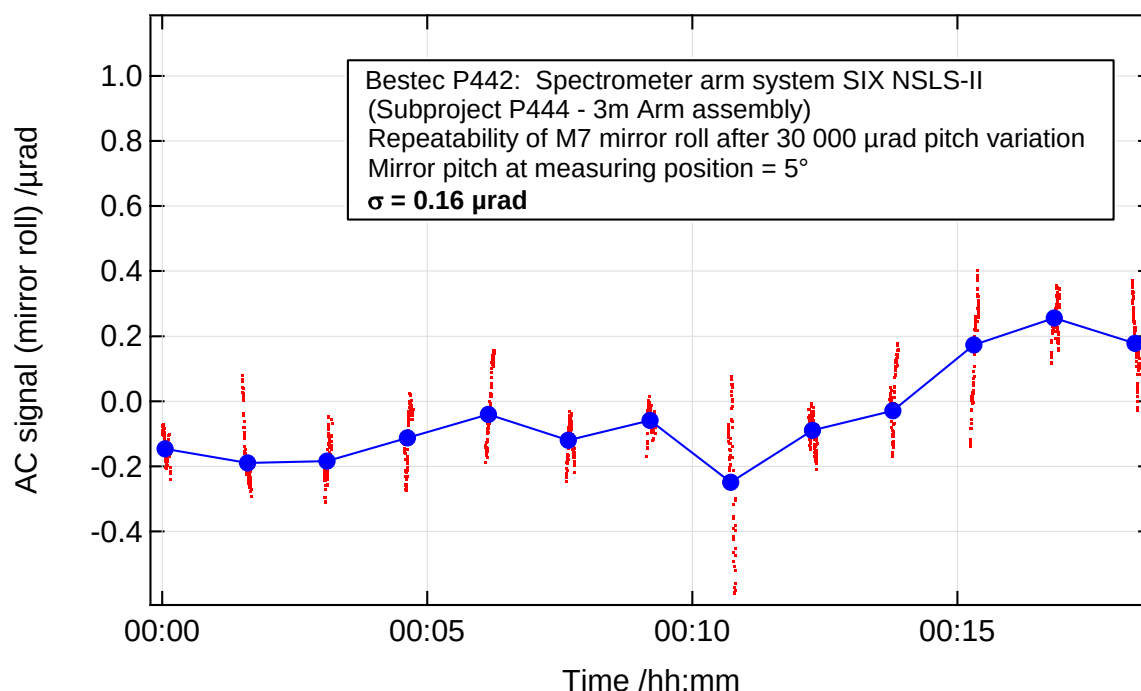


Figure 13: Autocollimator raw data and processed data showing the repeatability of M7 mirror roll upon mirror pitch variation (5° measuring position).

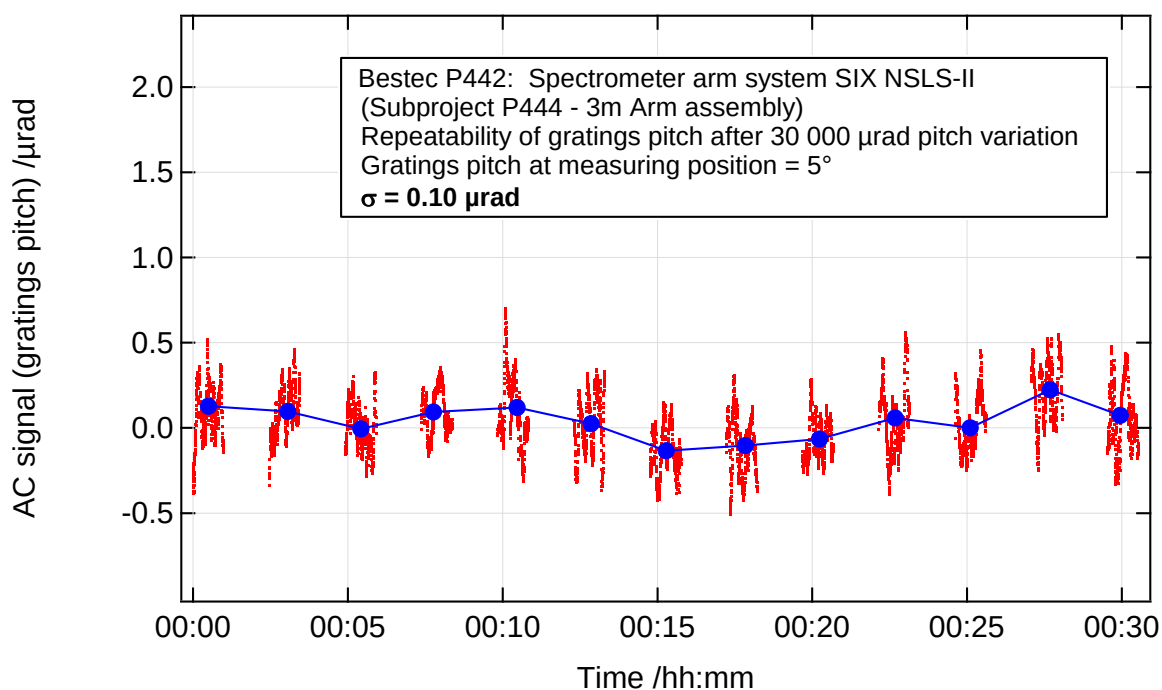
Repeatability of the gratings rotation (pitch, roll) upon gratings pitch variation

Figure 14: Autocollimator raw data and processed data showing the repeatability gratings pitch upon gratings pitch variation (5° measuring position).

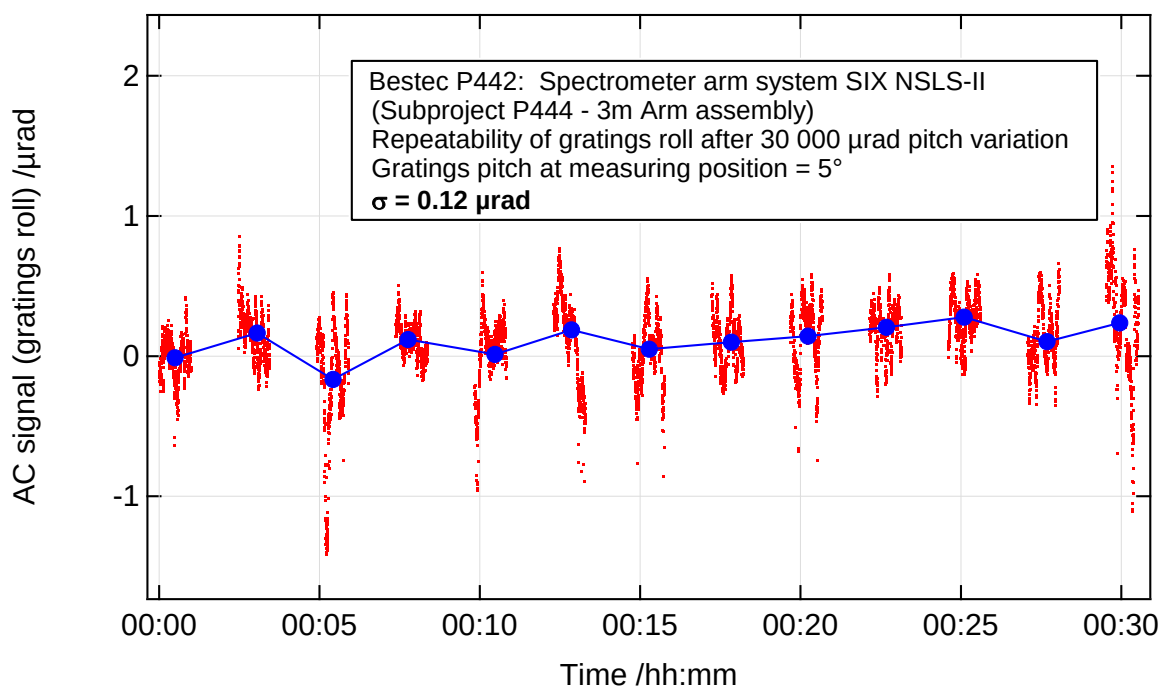


Figure 15: Autocollimator raw data and processed data showing the repeatability gratings roll upon gratings pitch variation (5° measuring position).

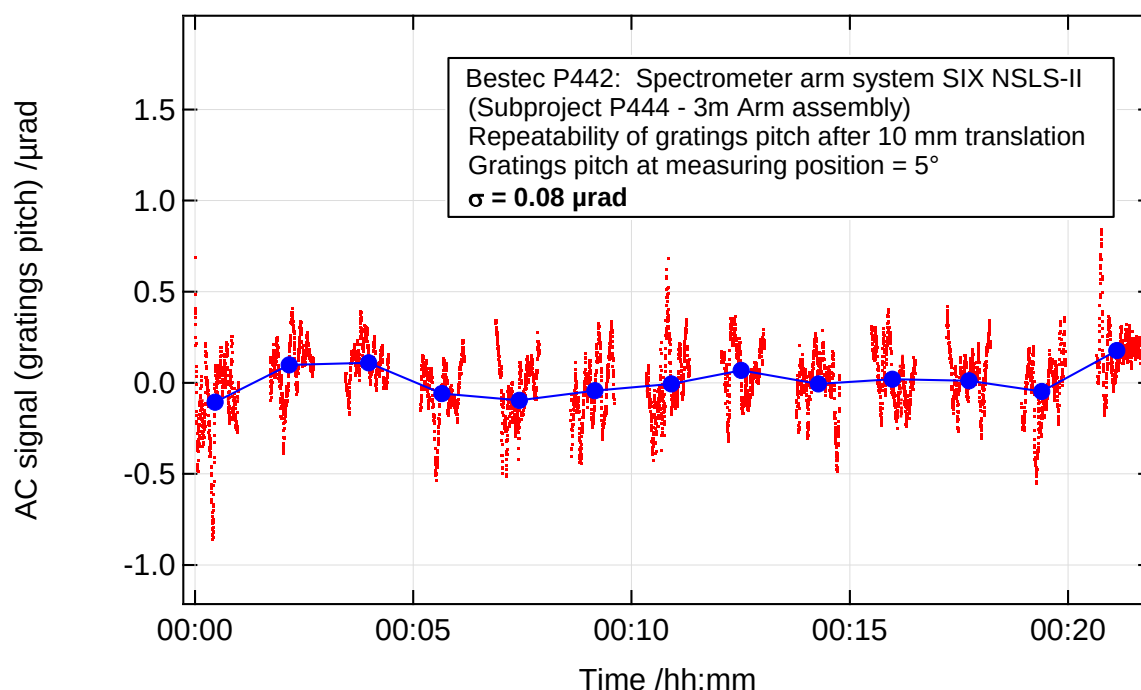
Repeatability of the gratings translation (pitch, roll) upon gratings translation

Figure 16: Autocollimator raw data and processed data showing the repeatability gratings pitch upon gratings translation (5° measuring position).

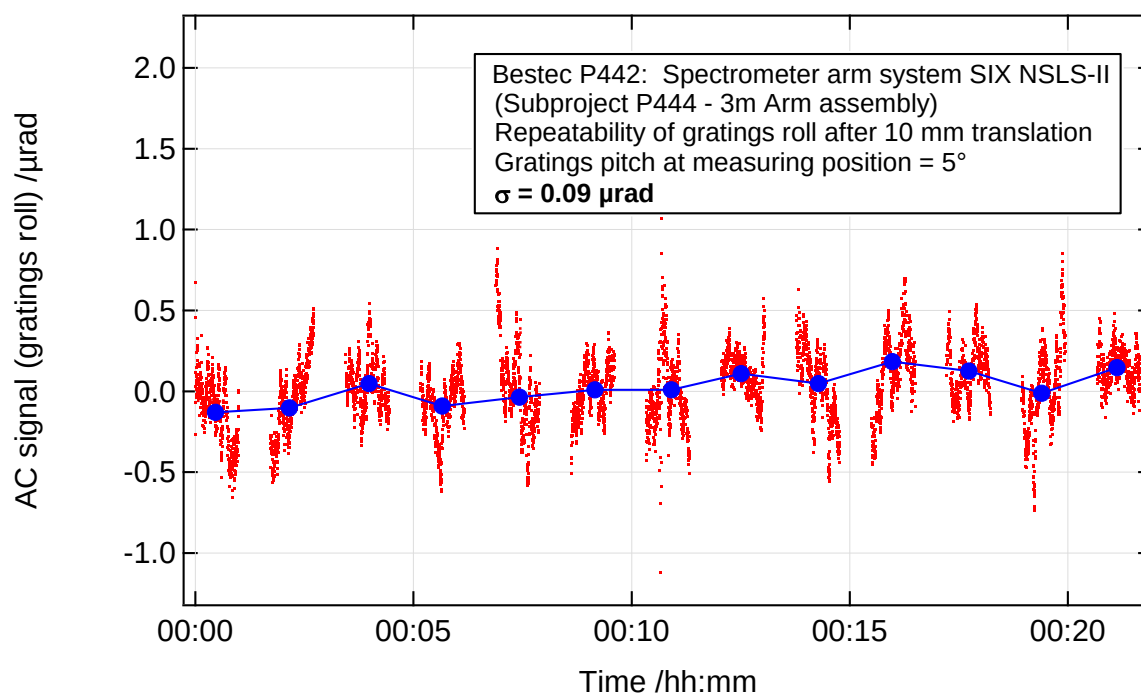


Figure 17: Autocollimator raw data and processed data showing the repeatability gratings roll upon gratings translation (5° measuring position).

Mirror M6 pitch positioning and long term stability scans

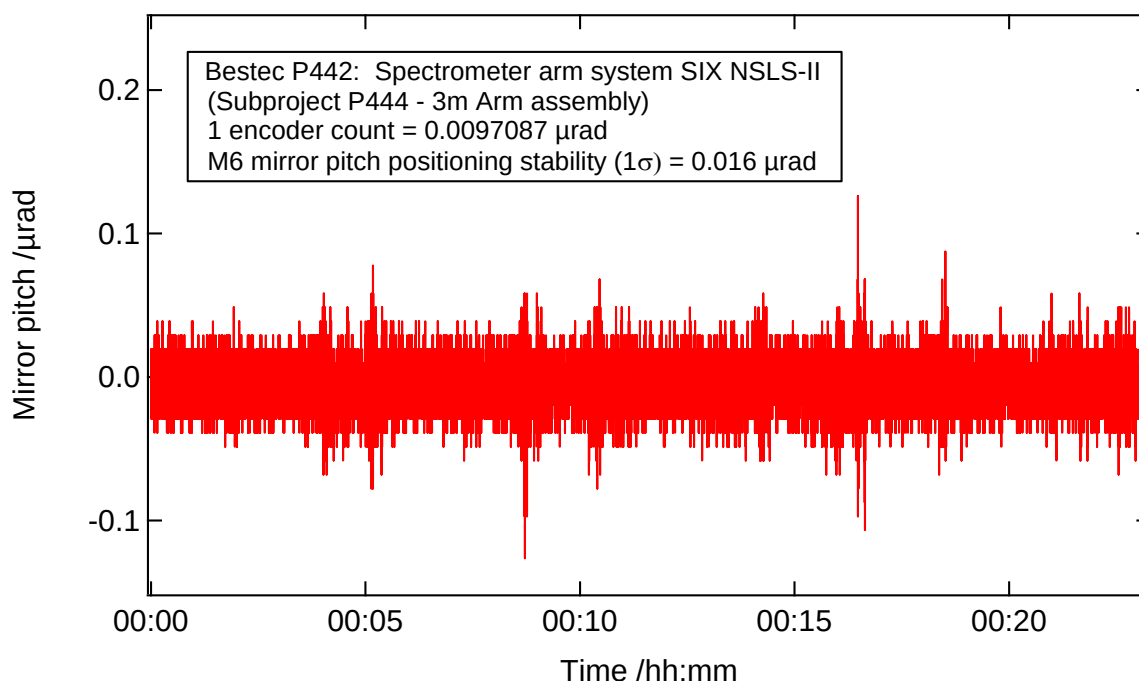


Figure 18: Long term M6 mirror pitch stability. Encoder data are neither averaged nor filtered.

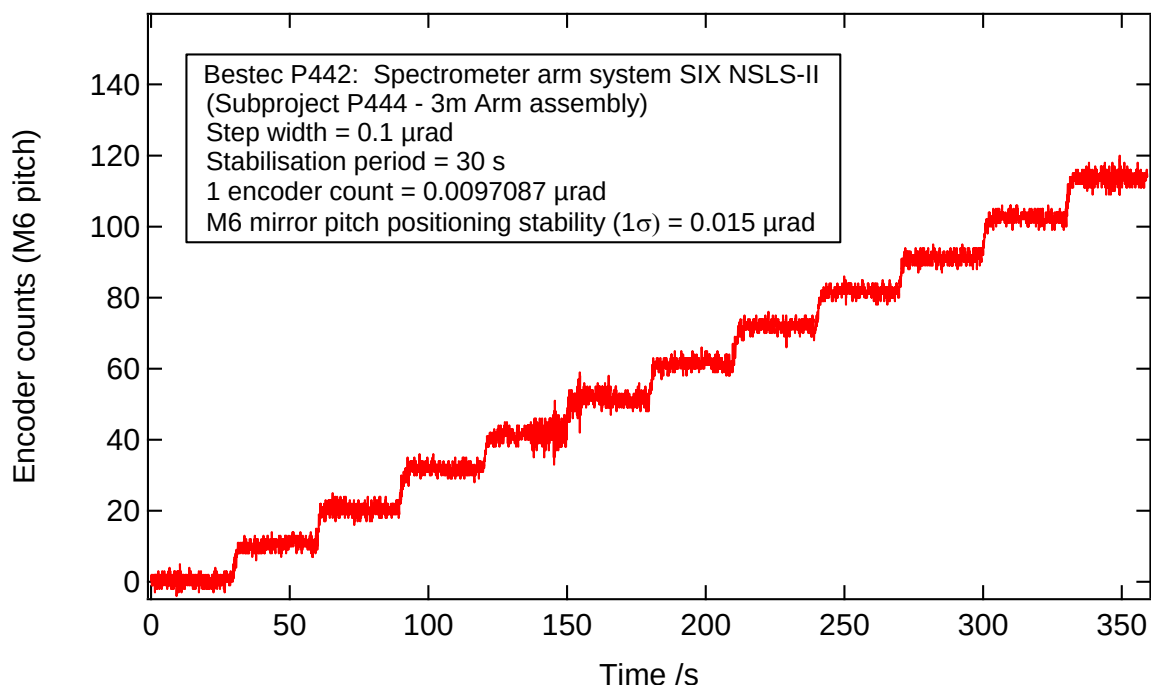


Figure 19: M6 mirror rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

Mirror M7 pitch positioning and long term stability scans

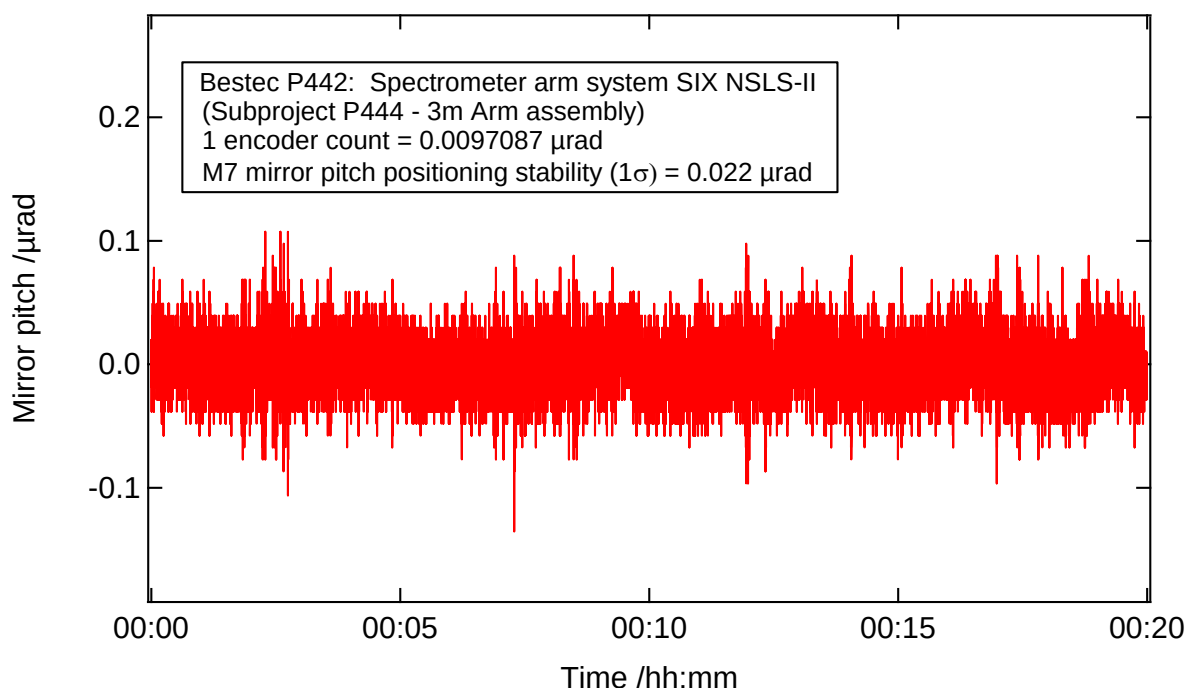


Figure 20: Long term M7 mirror pitch stability. Encoder data are neither averaged nor filtered.

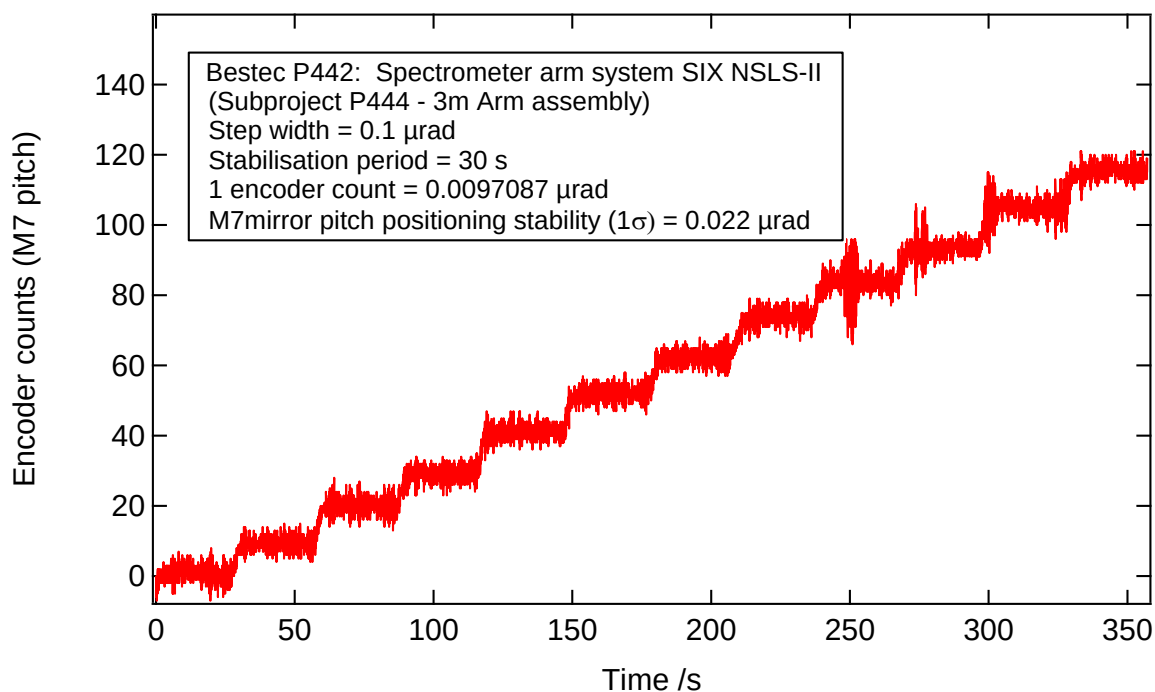


Figure 21: M7 mirror rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

Gratings pitch positioning and long term stability scans

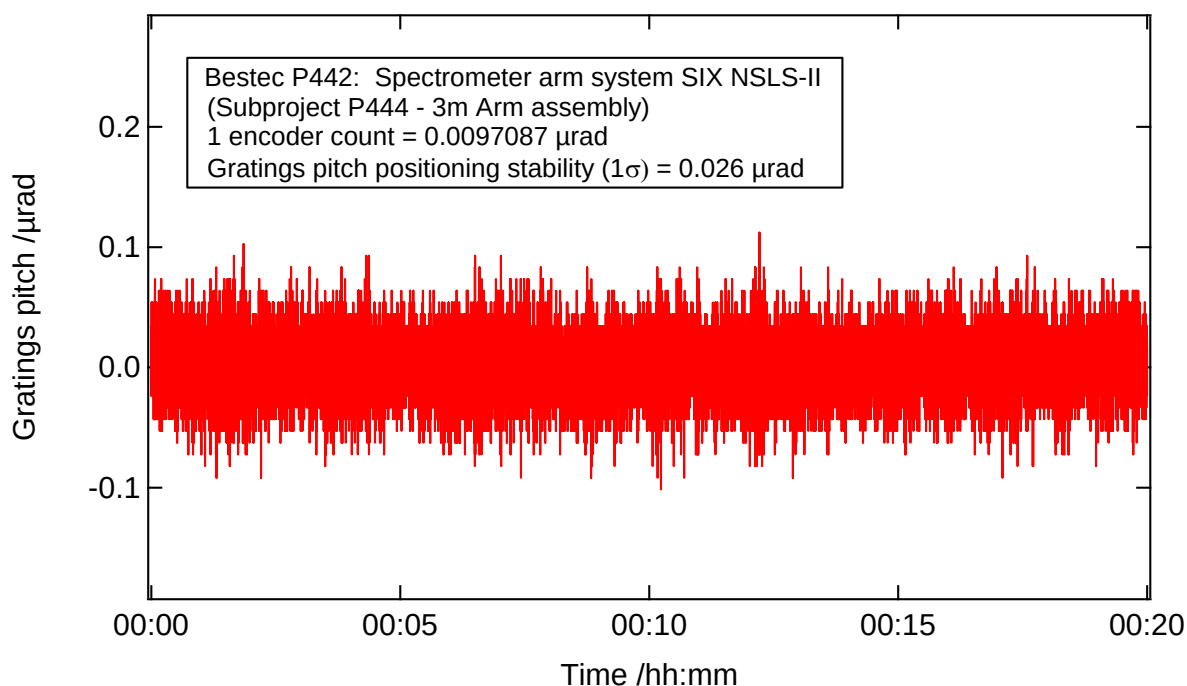


Figure 22: Long term gratings pitch stability. Encoder data are neither averaged nor filtered.

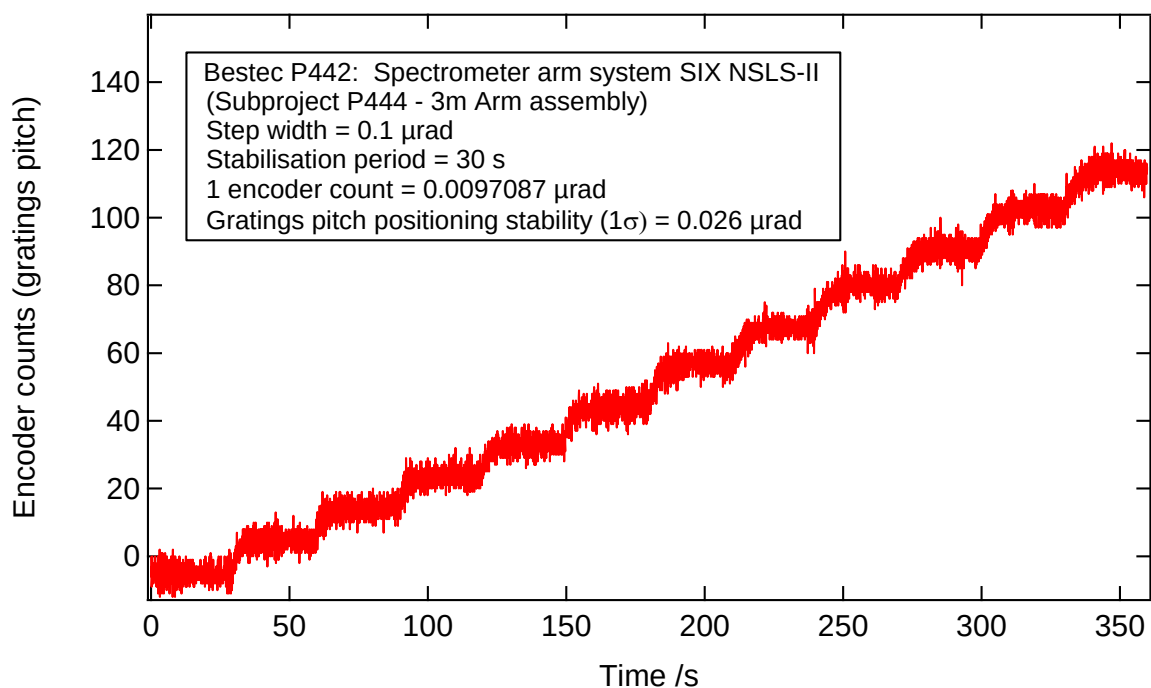


Figure 23: Gratings rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

Vacuum performance

The monochromator chamber was leak tested and baked up to 160°C. A sufficiently sensitive mass spectrometer (SRS RGA100) was installed at the vacuum chambers in order to monitor the partial pressures of the residual gas during and after bakeout procedure. A final pressure of 3.5×10^{-10} mbar was achieved.

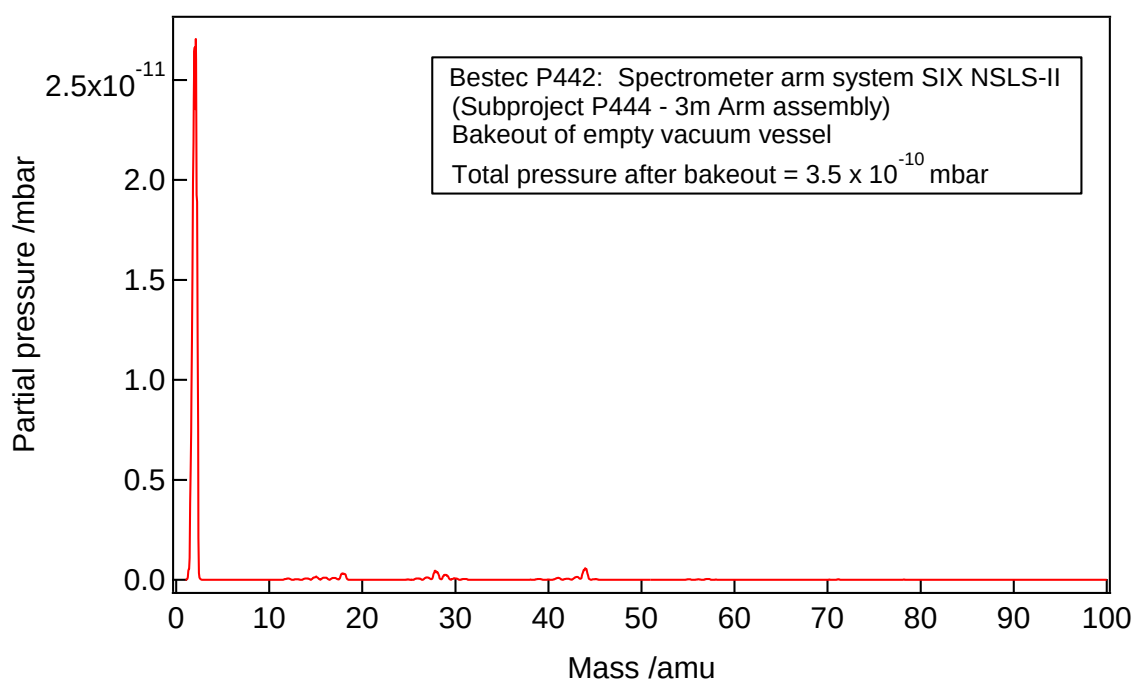


Figure 24: Vacuum of the empty monochromator chamber after bakeout.

2.3 DAA - detector arm and stand assembly

2.3.1 Test procedure

The repositioning of DAA was tested by using a laser tracker which was reading the position of a retroreflector, which was temporarily fixed on the detector radius traveler.

2.3.2 Technical specifications

Table 4: Overview of technical specifications (DAA)

General	
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	cylindrical, vertical axis
Main flange (chamber lid)	COF-type, diameter 700 mm
DAA height upstream left (DAAHUL)	
Range	± 4 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~ 0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height upstream right (DAAHUR)	
Range	± 4 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~ 0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height downstream (DAAHD)	
Range	± 4 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50

Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	$\sim 0.244 \mu\text{m}$ /count
Limit switches	2 x Schaltbau S880
2-theta rotation (DAAR2T)	
Motor	Phytron ZSH87/3.200.5-PLE80/20
Step resolution (full step)	$3.8957 \times 10^{-4} \text{ }^\circ$ /step
Encoder	Balluff Encoder scale: BML-M02-A55-A0-M3300 Readhead: BML-S1G0-B7ED-M5EZ-90-S284
Encoder resolution	$3.95668 \times 10^{-6} \text{ }^\circ$ /count
Limit switches	2 x Schaltbau S880
Detector translation (DT)	
Range	$\pm 250 \text{ mm}$
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	5 mm
Linear drive	HIWIN HGW35CC3R1220ZAPSS
Step resolution (full step)	0.5 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B10A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Aperture unit (AT, AB, AL, AR)	
Blade material	OFHC
Linear feedthrough	VAb MLS40-50
Stroke	50 mm
Motor	Phytron ZSS57.200.2,5-I with encoder
Spindle pitch	2 mm
Step resolution (full step)	10 μm /step at linear actuator

Limit switches

2 x Schaltbau S880

2.3.3 Test results

DAA Support repositioning

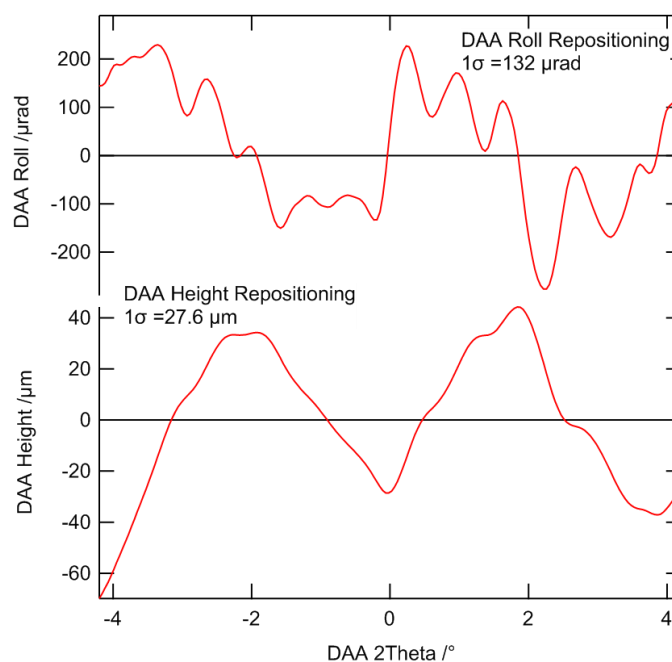


Figure 25: Roll and height repositioning X of the detector unit tested over the 10° DAA 2Theta arc available at Bestec.

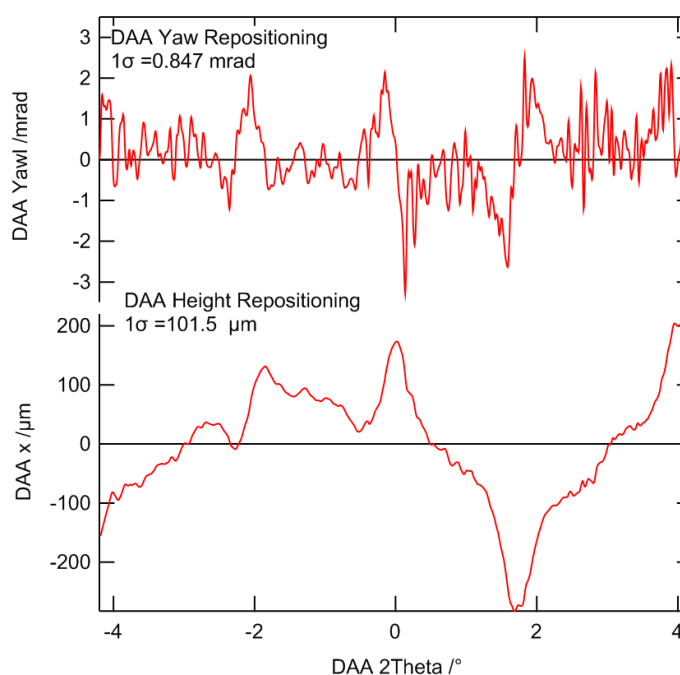


Figure 26: Yaw and repositioning X of the detector unit tested over the 10° DAA 2Theta arc available at Bestec.

Vacuum performance

The detector chamber was leak tested and baked up to 120°C. A sufficiently sensitive mass spectrometer (SRS RGA200) was installed at the vacuum chambers in order to monitor the partial pressures of the residual gas during and after bakeout procedure. A final pressure of 2.9×10^{-10} mbar was achieved.

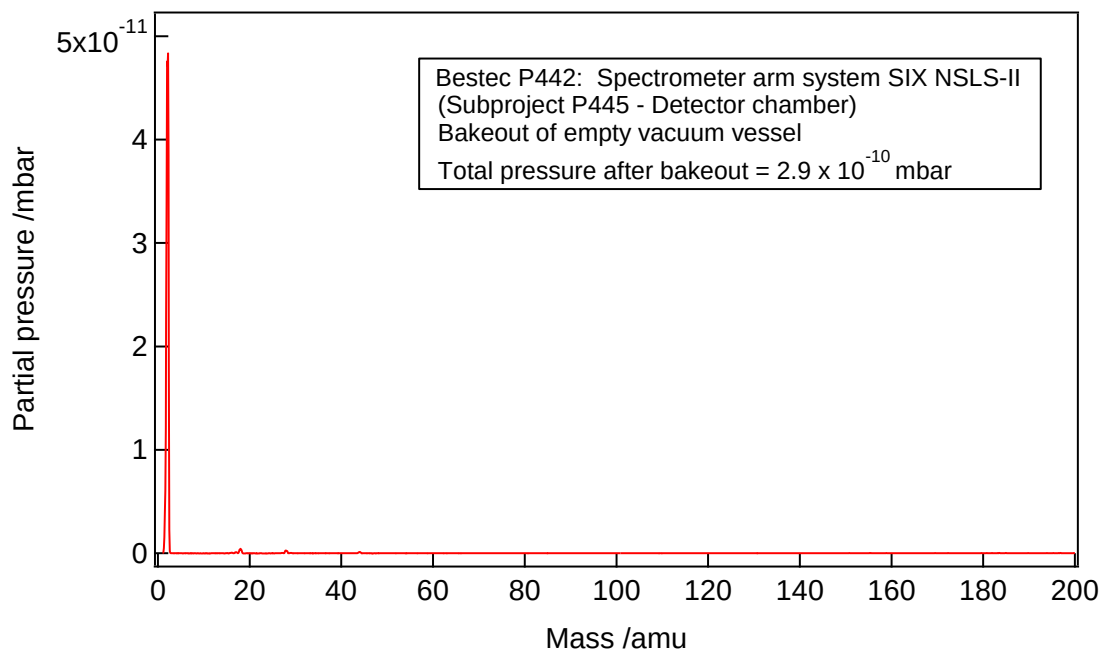


Figure 27: Vacuum of the empty detector chamber after bakeout.

2.4 System alignment

2.4.1 Test procedure

The 2Theta circles of M5, 3AA and DAA were aligned with respect to each other using a Laser tracker as diagnostics. The alignment is limited by the precision of the laser tracker. For DAA the small angle range available for the test leads to a strong coupling between the radius and the x and z coordinate parameters.



Figure 28: Spectrometer arm system.

2.4.2 Test results

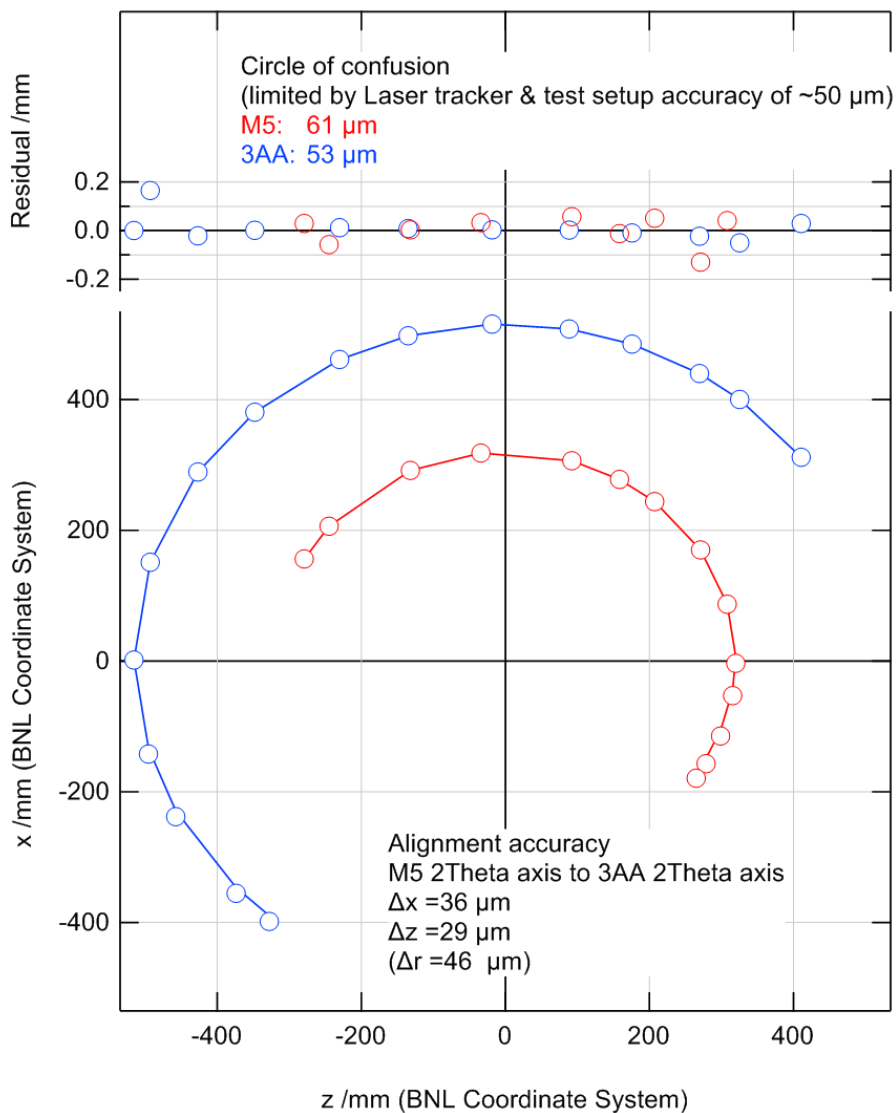


Figure 29: Alignment of the M5 and the 3AA axis.

The 3AA 2Theta and the M5 2Theta circles are coaxial to each other with an alignment error of $\Delta r = 46 \mu\text{m}$, which is limited by the laser tracker accuracy of $\sim 50 \mu\text{m}$.

3 Installation of the system at BNL - OSAT

The results summarized in this chapter were obtained during installation of the spectrometer arm system at the BNL.

3.1 M5 mirror system

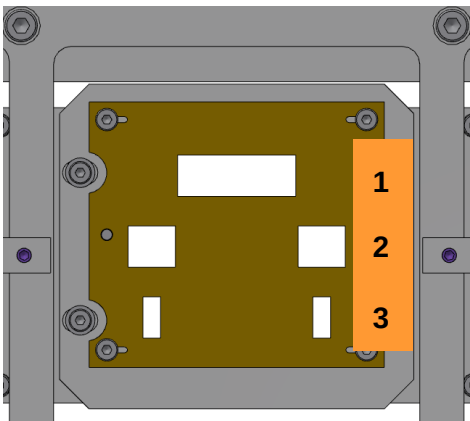
3.1.1 Technical specifications

Table 5: Overview of technical specifications (M5)

General	
Mirror unit	Mirror mechanics (tripod) included in the sample chamber
Beam height	1419.25 mm
Number of mirrors	2 collimating mirrors (M5A, M5B)
Substrate dimensions (L x W x H) Active optical surface (L x W)	260 mm x 50 mm x 72.5 mm 250 mm x 10 mm
Grazing angle	88.5°
M5 mirror rotation 2-theta (M5R2T)	
Total range	365°
Motor	Phytron ZSS57.200.2,5-GPL52/50
Step resolution (full step)	3.6×10^{-3} °/step
Encoder	Renishaw Resolute Encoder scale: RESA30USA350B Readhead: RA26BVA350B15V
Encoder resolution	5.36441×10^{-6} °/count
Limit switches	Huber LS01
FWD LSW-M5R2T	301710645 counts
REV LSW-M5R2T	- 4026308555 counts
M5 2Theta circle Offset from origin point	x = 29 µm z = 80 µm
Circle of minimum confusion	Max. radius = 1.86 µm RMS radius = 1.22 µm
Tangential wobble of the system	~8.7 µrad
M5 mirror pitch	
Range	± 15 mrad
Resolution	1 µrad

Repeatability of mirror rotation upon pitch variation (CLC) - FAT	Pitch: 0.42 μ rad Roll: 0.2 μ rad
Stability	1 μ rad
M5 mirror roll	
Range	± 15 mrad
Resolution	2 μ rad
Repeatability	10 μ rad
Stability	2 μ rad
M5 mirror yaw	
Range	± 20 mrad
Resolution	100 μ rad
Repeatability	500 μ rad
Stability	500 μ rad
M5 mirror lateral translation	
Range	± 5 mm
Resolution	0.1 μ m
Repeatability	0.5 μ m
Stability	1 μ m
M5 mirror longitudinal translation	
Range	± 5 mm
Resolution	0.1 μ m
Repeatability	0.5 μ m
Stability	1 μ m
Actuator drives for all six tripod axes (M5T1 -> M5T6)	
Motor	PI PiezoMike N-470.42U
Offset M5T1 (middle position)	101019622 counts
Offset M5T2 (middle position)	87469900 counts
Offset M5T3 (middle position)	139007368 counts
Offset M5T4 (middle position)	126941347 counts
Offset M5T5 (middle position)	60067971 counts
Offset M5T6 (middle position)	47245379 counts
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Encoder	Renishaw Resolute

	Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B50V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
FWD LSW-M5T1	103111142 counts
REV LSW-M5T1	98329867 counts
FWD LSW-M5T2	89759054 counts
REV LSW-M5T2	85897896 counts
FWD LSW-M5T3	141295076 counts
REV LSW-M5T3	136728375 counts
FWD LSW-M5T4	129236360 counts
REV LSW-M5T4	124391070 counts
FWD LSW-M5T5	62073690 counts
REV LSW-M5T5	57963734 counts
FWD LSW-M5T6	49482054 counts
REV LSW-M5T6	44979978 counts
Pivot translation X (PTX)	
Range	± 5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Pivot translation Z (PTZ)	
Range	± 5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S

	Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
M5 masks (M5M)	
Total translation range	0 – 55.2 mm
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1 mm
Step resolution (full step)	5 μm /step
Microstepping 1/64	0.078 μm /step
Limit switches	2 x Huber LS01
REV LSW-M5M	0
FWD LSW-M5M	707418 steps
	Offset (1) = 87880 steps Offset (2) = 241410 steps Offset (3) = 395630 steps (middle positions of the apertures)
Motor retractable photodiode (MPD3)	
Motor	PI PiezoMike N-470.42U
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Limit switches	2 x Huber LS01
Diagnostic slits	
Alignment accuracy (factory-aligned)	3 μm

3.1.2 Test results

M5 2Theta circle

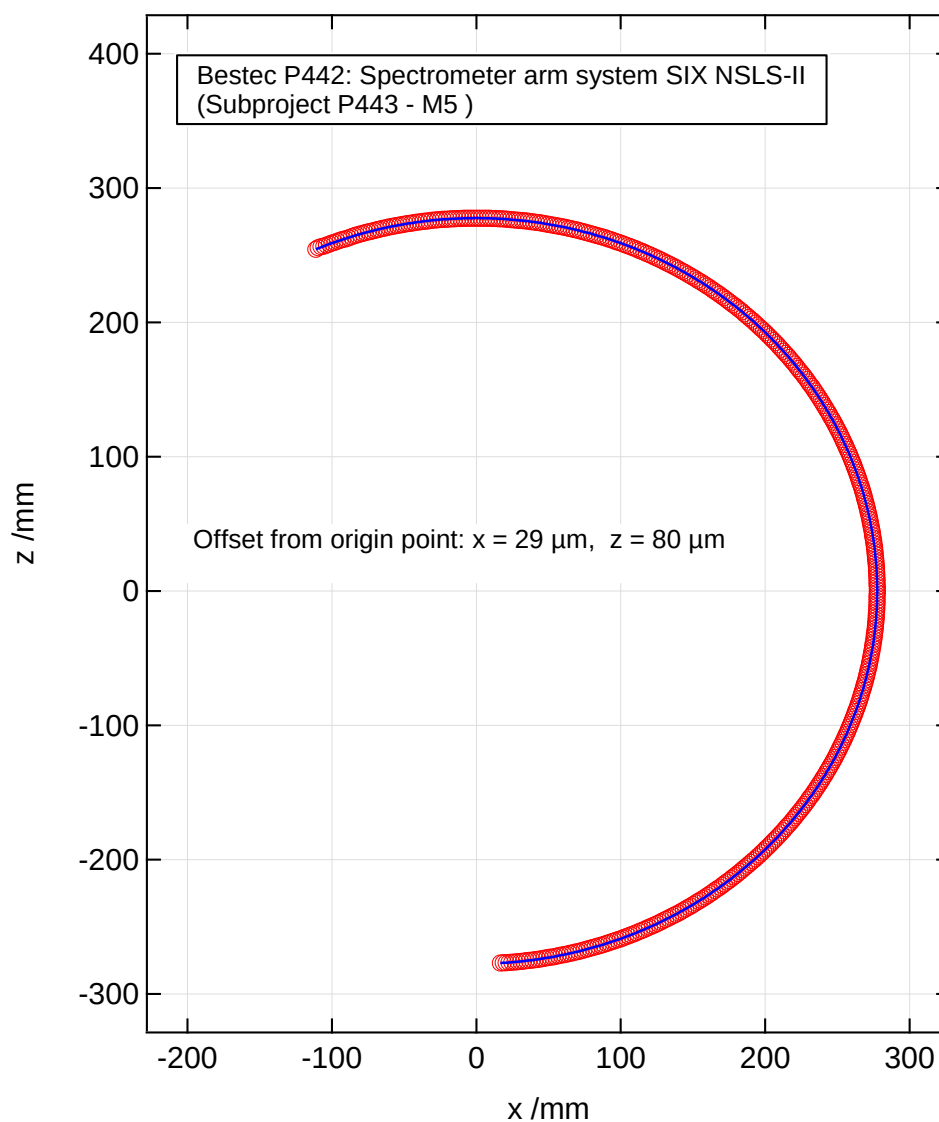


Figure 30: M5 2Theta circle.

Circle of minimum confusion of M5 2Theta

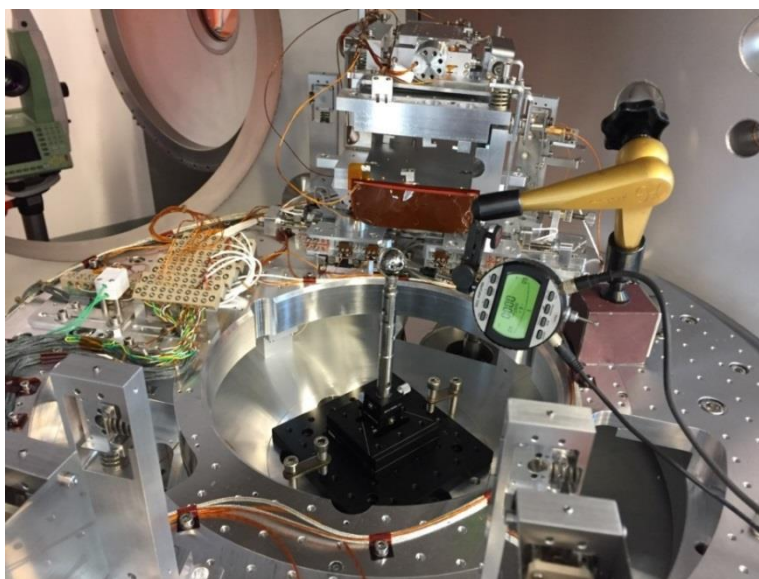


Figure 31: Test setup for the determination of the circle of minimum confusion of the M5 2Theta rotation stage.

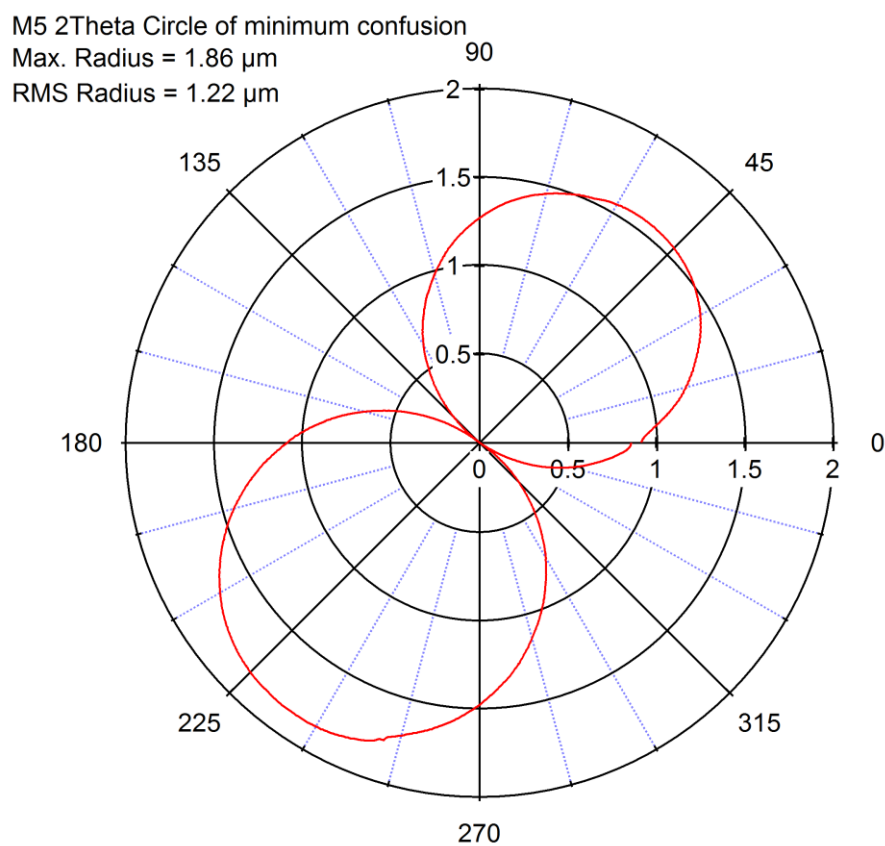


Figure 32: Circle of minimum confusion of the M5 2Theta rotation stage.

Wobble of M5 2Theta

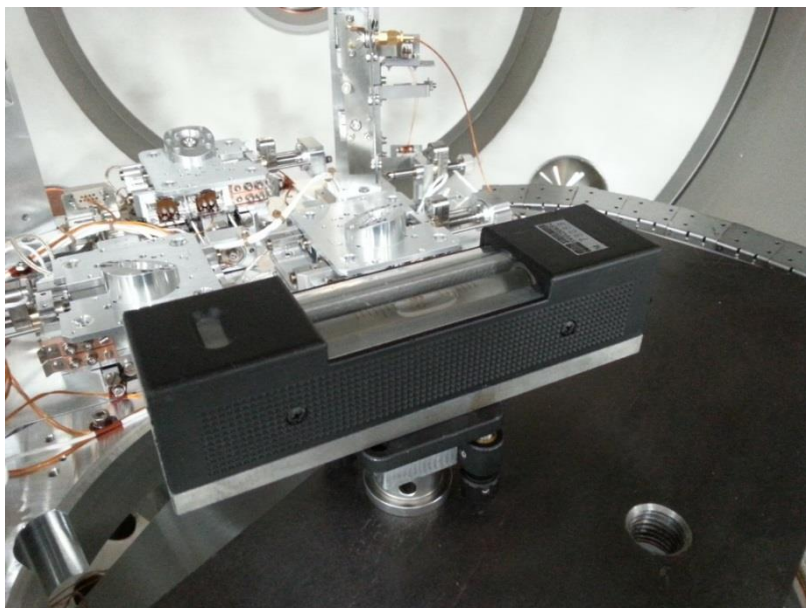


Figure 33: Test setup for the determination of the tangential wobble of the M5 2Theta rotation stage.

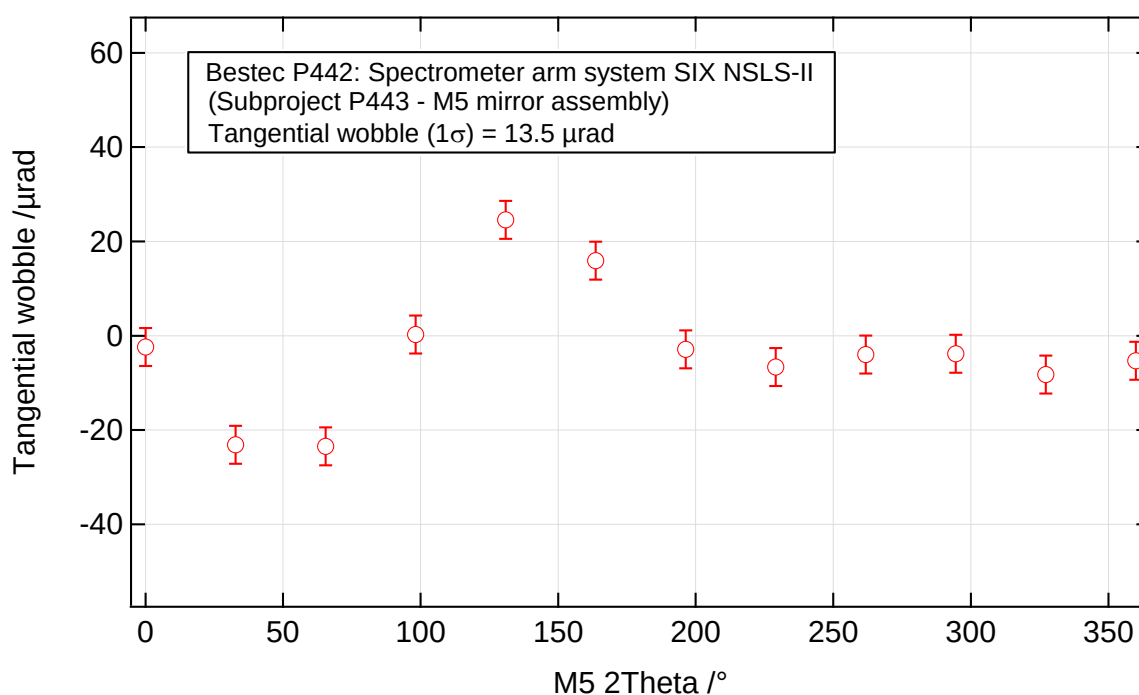


Figure 34: Tangential wobble of the M5 2Theta rotation stage.

Taking into account the uncertainty of the water level reading of $\sim 10 \mu\text{rad}$ the tangential wobble of the system only is $\sim 8.7 \mu\text{rad}$.

3.1.3 Additional hardware tests

Tests of the electrical components

Table 6: Overview of tested electrical components

Hardware abbreviation	Test method	Status
TC1	Temperature reading	OK
TC2	Temperature reading	OK
TC3	Temperature reading	OK
TC4	Temperature reading	OK
TC5	Temperature reading	OK
TC6	Temperature reading	OK
CSW-M5T1	Resistance	OK
PD1	Diode characteristic 0.5 V	OK
PD2	Diode characteristic 0.5 V	OK
PD3	Diode characteristic 0.5 V	OK

3.1.4 Location of in-vacuum components and sensors

Table 7: Location of in-vacuum components and sensors

In air (feedthrough)	In vacuum	Wiring scheme
TC1	At the base plate of the upper mirror	443 - 7M - 346a
TC2	At the base plate of the lower mirror	443 - 7M - 346a
TC3	At 2Theta motor holder plate	443 - 7M - 346b
TC4	At one of the 2Theta trailing chain support plate	443 - 7M - 346b
TC5	Inboard side of 2Theta support pillar	443 - 7M - 346b
TC6	Outboard side of 2Theta support pillar	443 - 7M - 346b

3.2 3AA - 3m arm and optics tank assembly

3.2.1 Technical specifications

Table 8: Overview of technical specifications (M6)

M6 mirror rotation (M6P)	
Range relative to horizontal	- 3.51° to 1.26°
Offset (the optical surface is horizontal)	544192 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.09 μ rad Roll: 0.09 μ rad
M6 mirror pitch stability (CLC)	0.016 μ rad
Linear drive	PE2.4.0100GRT 12x1
Motor	Phytron ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	240
Sine-bar connecting rod	115 mm
Angle offset	35°
Step resolution (full step)	0.42 μ rad /step
Encoder	Renishaw Tonic Encoder scale: RGSZ20-S Readhead (2x): T2641-30M Interpolator (2x): TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-M6P	- 5759381 counts
FWD LSW-M6P	2806550 counts
M6 mirror translation (M6TZ)	
Range	± 10 mm
Repeatability of mirror rotation upon rotation (CLC) - FAT	Pitch: 0.08 μ rad Roll: 0.16 μ rad
M6 mirror pitch stability (CLC) - FAT	0.022 μ rad
Motor	PI PiezoMike N-470.42U
Travel range	26 mm

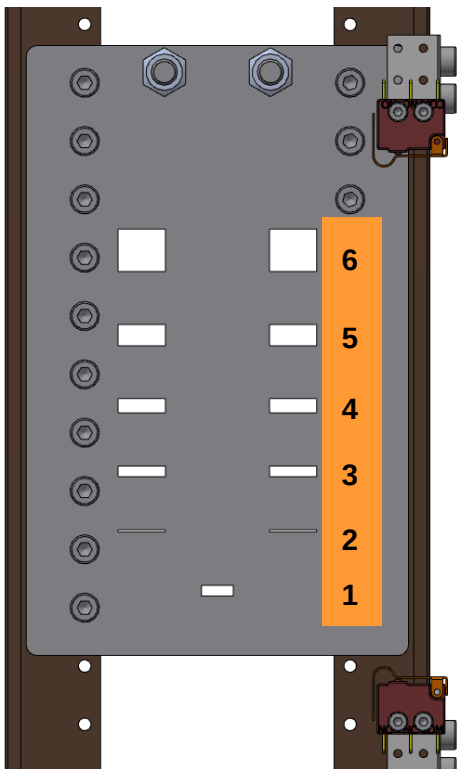
Typical step size	20 nm (max. 30 nm)
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
REV LSW-M6TZ	19600441 counts
FWD LSW-M6TZ	23642188 counts
M6 masks (M6M)	
Total translation range	0 – 90 mm
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1.25 mm
Step resolution (full step)	6.25 μm /step
Microstepping 1/64	0.098 μm /step
Limit switches	2 x Huber LS01
REV LSW-M6M	0
FWD LSW-M6M	917423 counts
 Offset (1) = 178000 steps Offset (2) = 296000 steps Offset (3) = 412000 steps Offset (4) = 540000 steps Offset (5) = 676000 steps Offset (6) = 844000 steps (middle positions of the apertures)	

Table 9: Overview of technical specifications (vacuum vessel, M7, gratings)

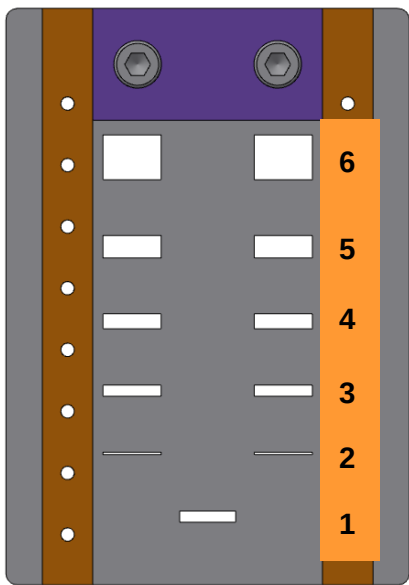
General	
Included components	M6 mirror mechanics M7 mirror mechanics Gratings mechanics M7 masks
Total weight of the system	~2000 kg (without lower granite block)
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	Stadium shaped
Main flange (chamber lid)	At the top (wire sealing)
Number of deflecting mirrors	1
Deflecting direction	Downward in vertical plane
Substrate dimensions (L x W x H) Active optical surface (L x W)	540 mm x 60 mm x 70 mm 525 mm x 40 mm
Number of gratings	3
Substrate dimensions (L x W x H) Active optical surface (L x W)	220 mm x 55 mm x 50 mm 200 mm x 40 mm
Base pressure of empty chamber after bakeout	3.5×10^{-10} mbar
M7 mirror and grating axis position (Reimer-Torge parameters)	
Beam offset	20 mm
Horizontal position of the plane mirror axis with respect to the grating axis	0.53 mm (closer to the source)
Vertical position of the plane mirror axis with respect to the grating axis	10.072 mm (below grating rotation axis)
Radius of rotation axis	20.0856 mm
Distance between mirror and mirror axis	61 mm
M7 mirror rotation (M7P)	
Range relative to horizontal	- 12.03° to 1.31°
Offset (the optical surface is horizontal)	475859 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.16 μ rad Roll: 0.09 μ rad
M7 mirror pitch stability (CLC)	0.029 μ rad
Linear drive	ERO MOLS 100.100

	(with threaded spindle 12 x 1)
Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	495 mm
Sine-bar connecting rod	115 mm
Angle	15°
Step resolution (full step)	0.20 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206 Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-M7P	- 21162794 counts
FWD LSW-M7P	2831215 counts
Gratings rotation (GP)	
Range relative to horizontal	- 14.19° to 1.74°
Offset (the optical surface is horizontal)	- 1081502 counts
Repeatability of grating rotation upon rotation (CLS)	Pitch: 0.10 μ rad Roll: 0.09 μ rad
Repeatability of grating rotation upon translation (CLS)	Pitch: 0.17 μ rad Roll: 0.18 μ rad
Grating pitch stability	0.032 μ rad
Linear drive	PE2.4.0100GRT 12x1
Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	340 mm
Sine-bar connecting rod	115 mm
Angle offset	12°
Step resolution	0.3 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206

	Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Collision switches	2 x Huber LS01
REV LSW-GP	- 26588173 counts
FWD LSW-GP	2044496 counts
Gratings translation (GT)	
Range	0 mm to 175.85 mm
Offset (the middle grating is centered on the beam)	275620425 counts
Linear drive	Bestec linear drive
Motor	Phytron ZSS57.200.2,5
Spindle pitch	2 mm
Step resolution (full step) Belt gear 2:1	10 μ m /step 5 μ m /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μ m /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-GT	259900631 counts
FWD LSW-GT	295071198 counts
Optics base plate translation (BTZ)	
Range	- 11.5 mm to 11.5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μ m /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μ m /count

Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-BTZ	15092500 counts
FWD LSW-BTZ	14630600 counts
2-theta rotation (3AAR2T)	
Total range	112°
Offset (at this position the incident beam on M6 is perpendicular to the incident beam on the sample)	44605742 counts
Motor	ZSH87/3.200.5.PLE80-80:1
Step resolution (full step)	4.45686×10^{-4} °/step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B10A
Encoder resolution	4.98311×10^{-6} °/count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
Collision switch	4 x Schaltbau S880
REV LSW-3AAR2T	33278342 counts
FWD LSW-3AAR2T	55927802 counts
3AA 2Theta circle Offset from origin point	x = 1.9 µm z = 3.5 µm
Circle of minimum confusion of the 3AA 2Theta radial bearing	Max. radius = 27 µm RMS radius = 7.9 µm
Height deviation along 3AA 2Theta circle of the 3AA 2Theta radial bearing	6.4 µm
Circle of the minimum confusion of the 3AA 2Theta - coupled motion of 3AA and DAA	Max. radius = 135 µm RMS radius = 66 µm (inner circle) Max. radius = 155 µm RMS radius = 57 µm (outer circle)
Height deviation of the 3AA 2Theta - coupled motion of 3AA and DAA	14 µm (inner circle) 14.3 µm (outer circle)
Wedge leveler outboard (OVY)	
Range	- 5.19 mm to 5.12 mm
Offset ¹	19730405 counts

Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-OVY	19632356 counts
FWD LSW-OVY	19838414 counts
Wedge leveler inboard (IVY)	
Range	- 4.79 mm to 5.41 mm
Offset ¹	14680064 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-IVY	17622865 counts
FWD LSW-IVY	17826826 counts
M7 masks (M7M)	
Total translation range	0 – 96 mm
Motor	Phytron ZSS57.200.2,5
Linear drive	VAb LD40-100
Spindle pitch	3 mm
Step resolution (full step)	15 μm /step
Microstepping 1/64	0.234 μm /step
Limit switches	2 x Schaltbau 880
REV LSW-M7M	- 408989 steps

FWD LSW-M7M	0
	Offset (1) = -72000 steps Offset (2) = -119280 steps Offset (3) = -168285 steps Offset (4) = -220400 steps Offset (5) = -277170 steps Offset (6) = -346000 steps (middle positions of the apertures)

¹ When the offset positions of the OVY- and IVY-axes are simultaneously reached and the pitch of the optical elements is 0°, then the optical elements are at the nominal height and their roll is 0°.

3.2.2 Test results

Repeatability of the mirror M6 rotation (pitch, roll) upon mirror pitch variation

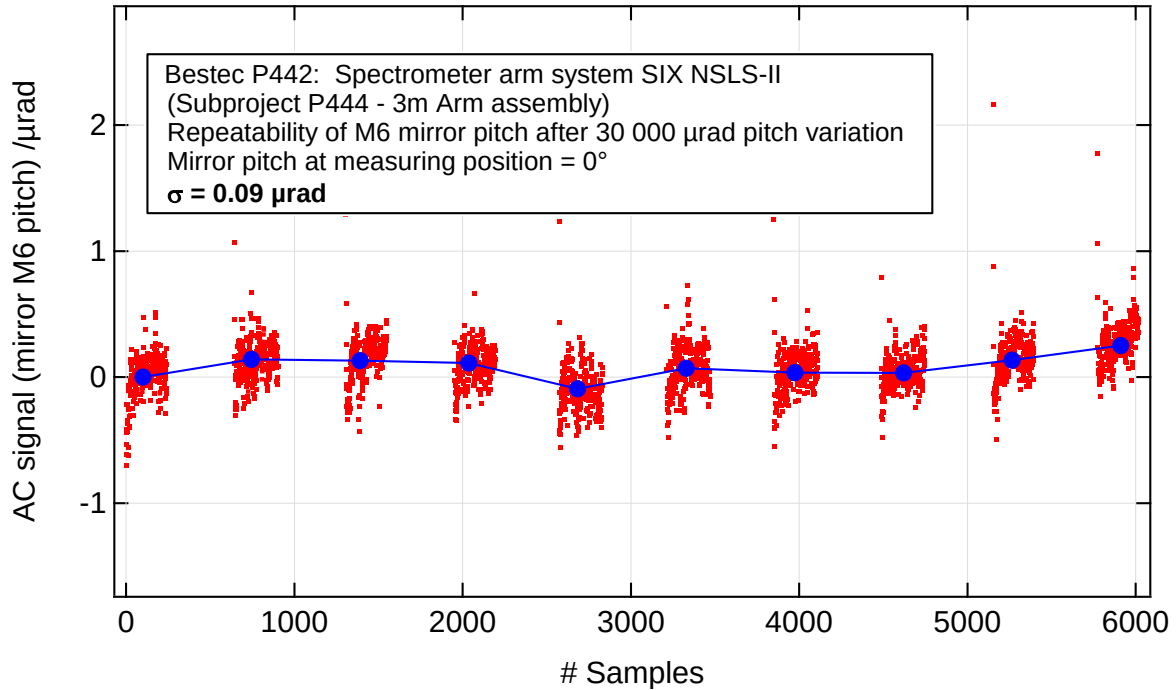


Figure 35: Autocollimator raw data and processed data showing the repeatability of M6 mirror pitch upon mirror pitch variation (0° measuring position).

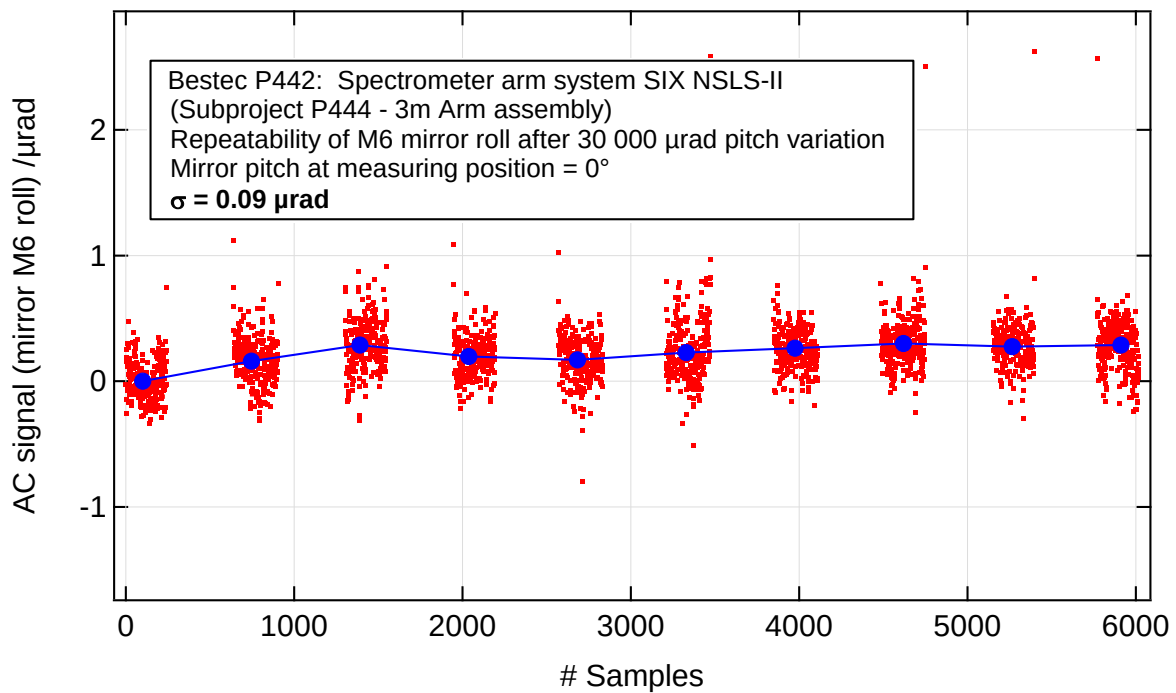


Figure 36: Autocollimator raw data and processed data showing the repeatability of M6 mirror roll upon mirror pitch variation (0° measuring position).

Repeatability of the mirror M7 rotation (pitch, roll) upon mirror pitch variation

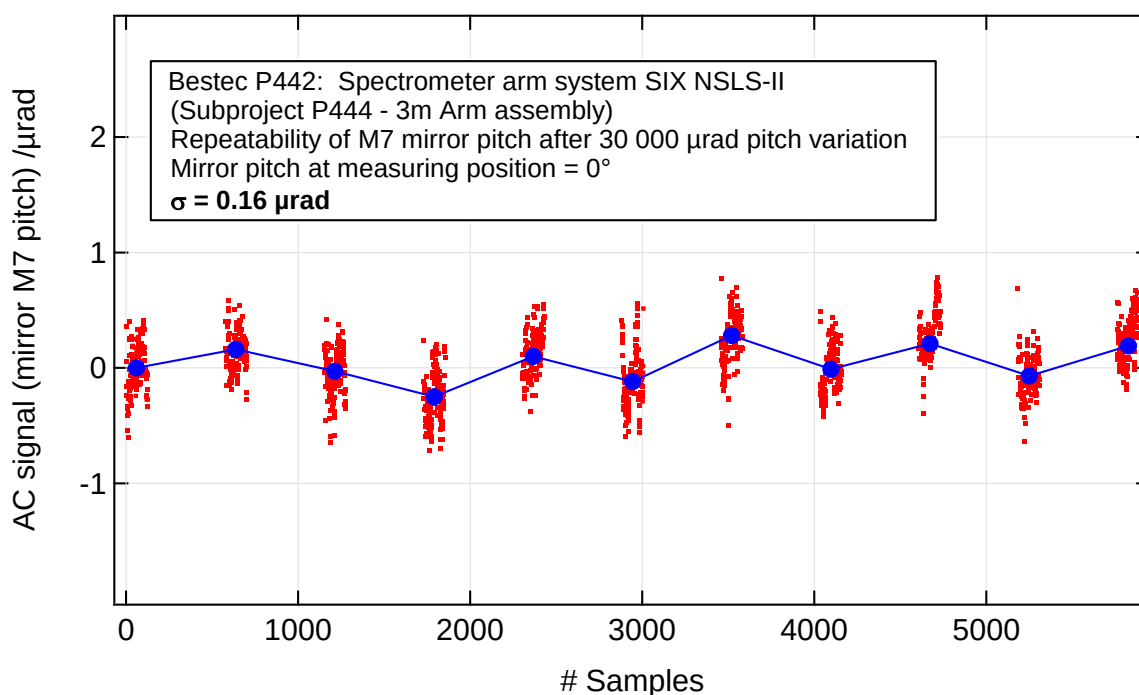


Figure 37: Autocollimator raw data and processed data showing the repeatability of M7 mirror pitch upon mirror pitch variation (0° measuring position).

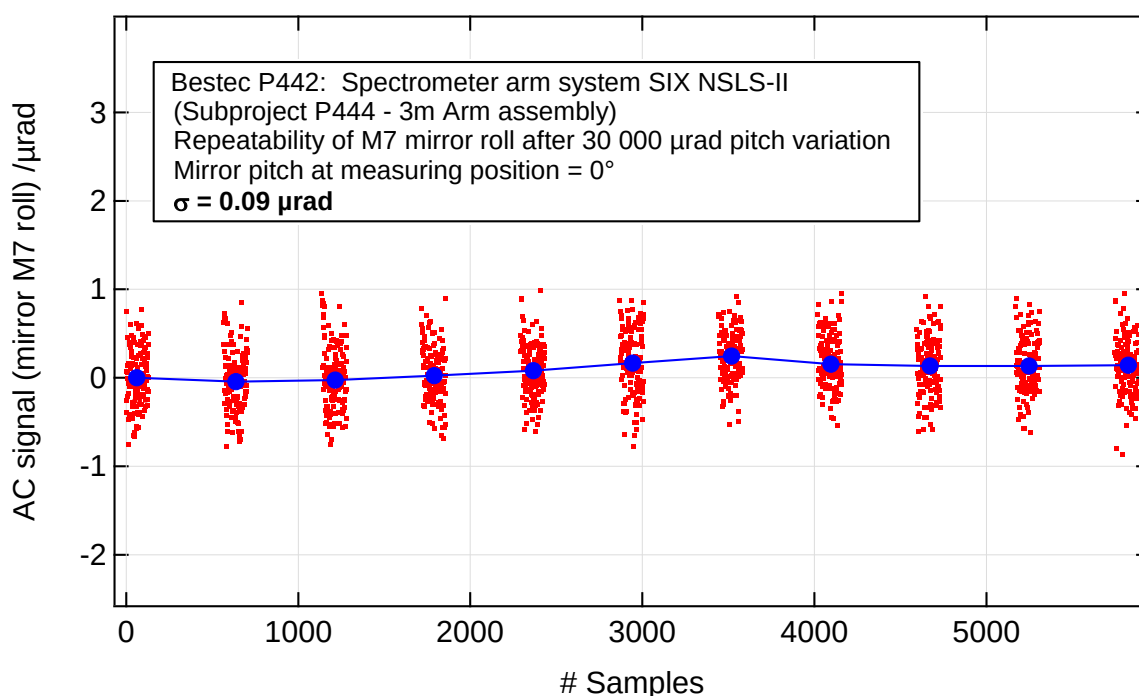


Figure 38: Autocollimator raw data and processed data showing the repeatability of M7 mirror roll upon mirror pitch variation (0° measuring position).

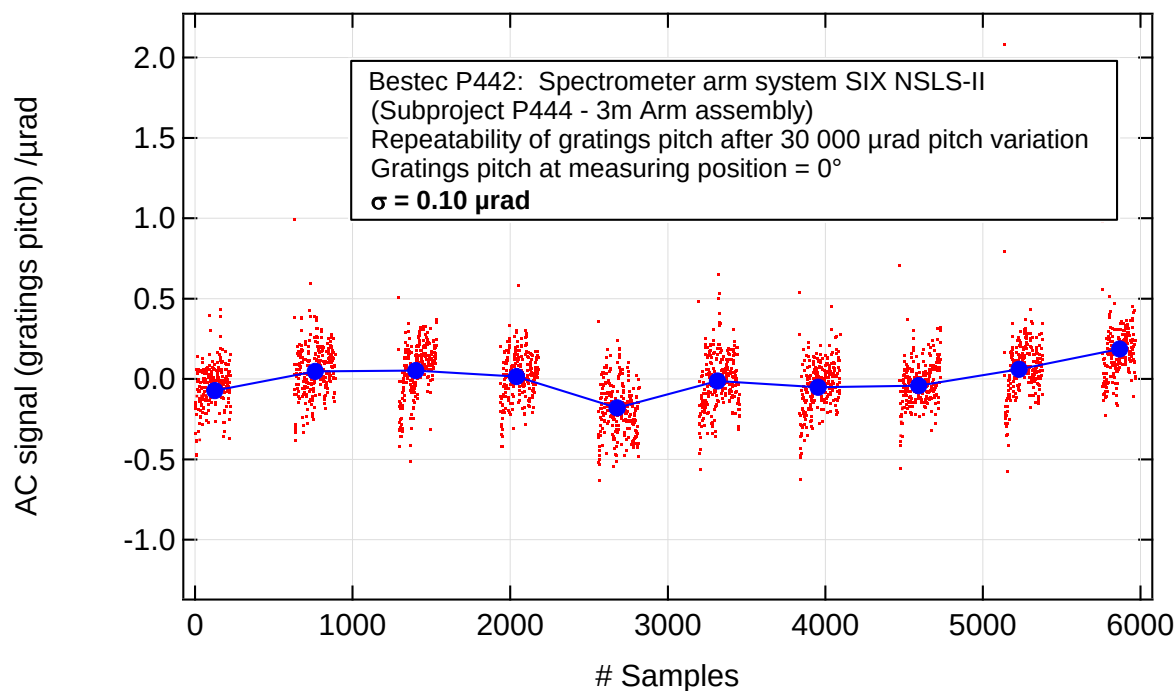
Repeatability of the gratings rotation (pitch, roll) upon gratings pitch variation

Figure 39: Autocollimator raw data and processed data showing the repeatability gratings pitch upon gratings pitch variation (0° measuring position).

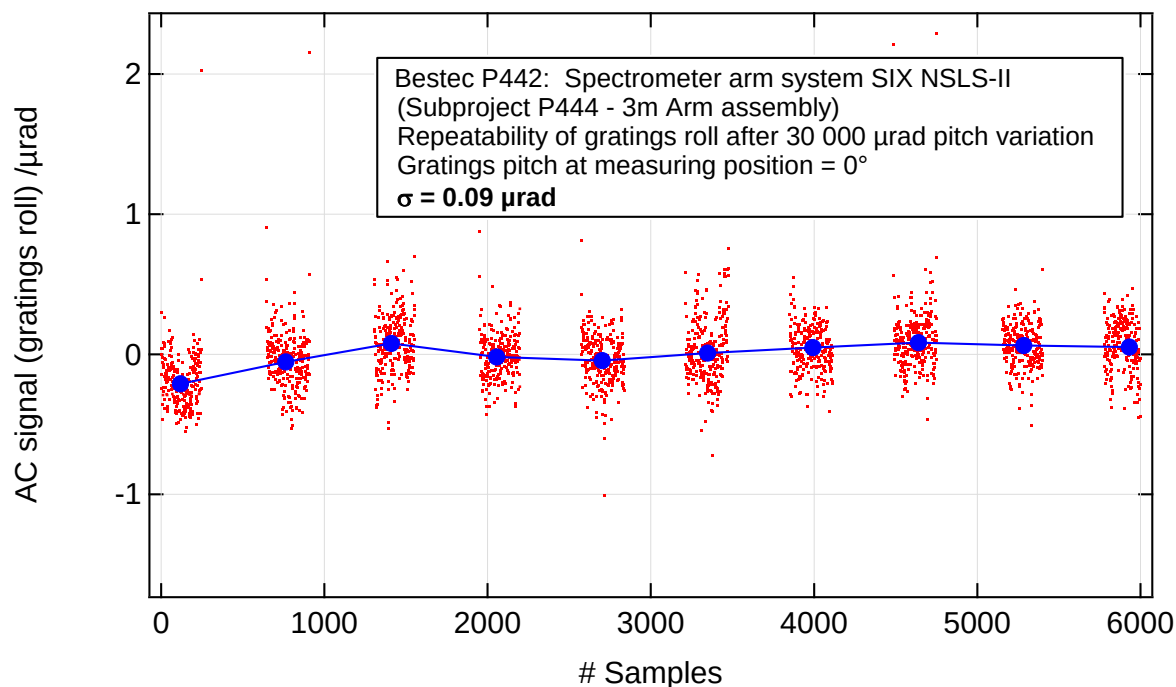


Figure 40: Autocollimator raw data and processed data showing the repeatability gratings roll upon gratings pitch variation (0° measuring position).

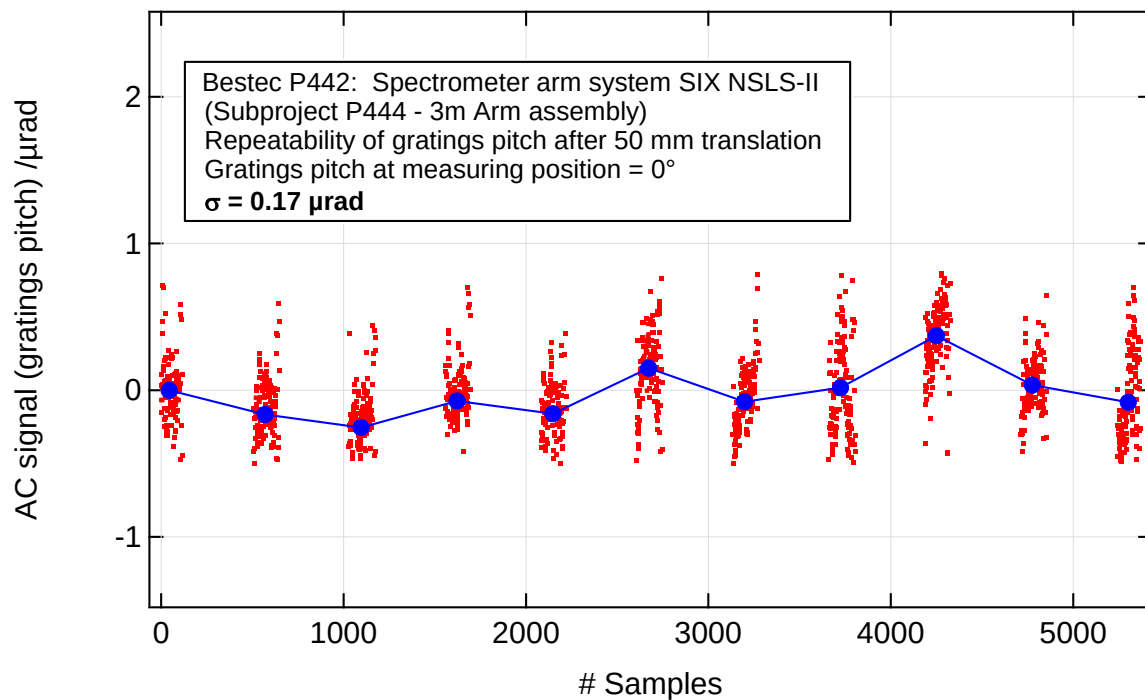
Repeatability of the gratings translation (pitch, roll) upon gratings translation

Figure 41: Autocollimator raw data and processed data showing the repeatability gratings pitch upon gratings translation (0° measuring position).

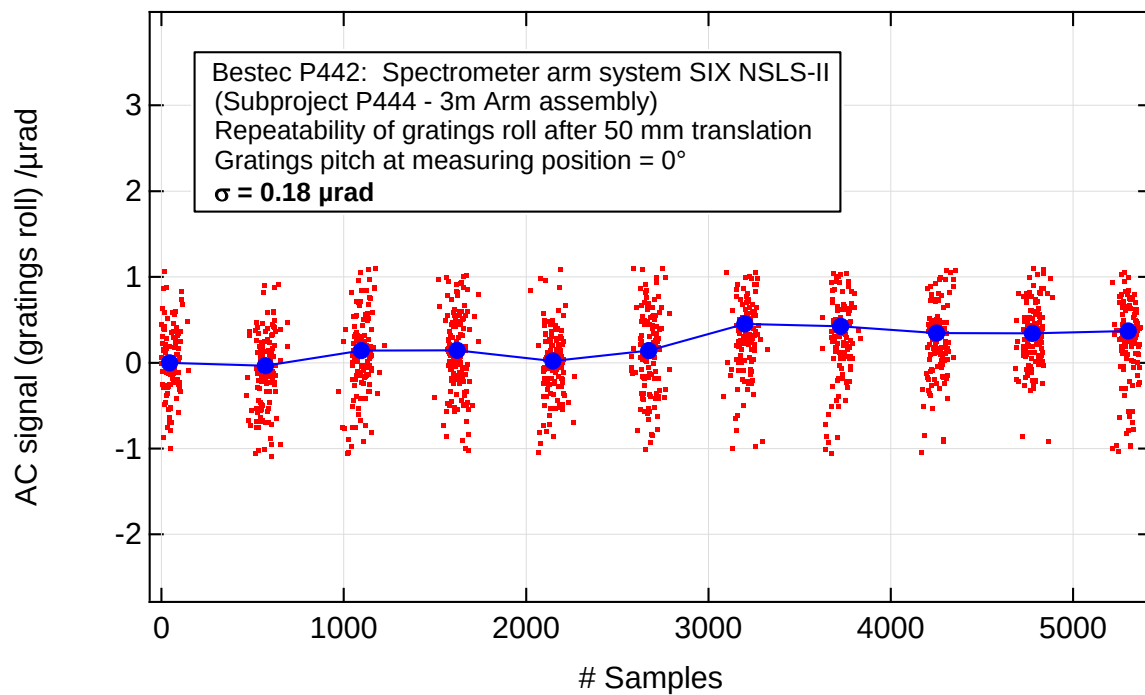


Figure 42: Autocollimator raw data and processed data showing the repeatability gratings roll upon gratings translation (0° measuring position).

Mirror M6 pitch positioning and long term stability scans

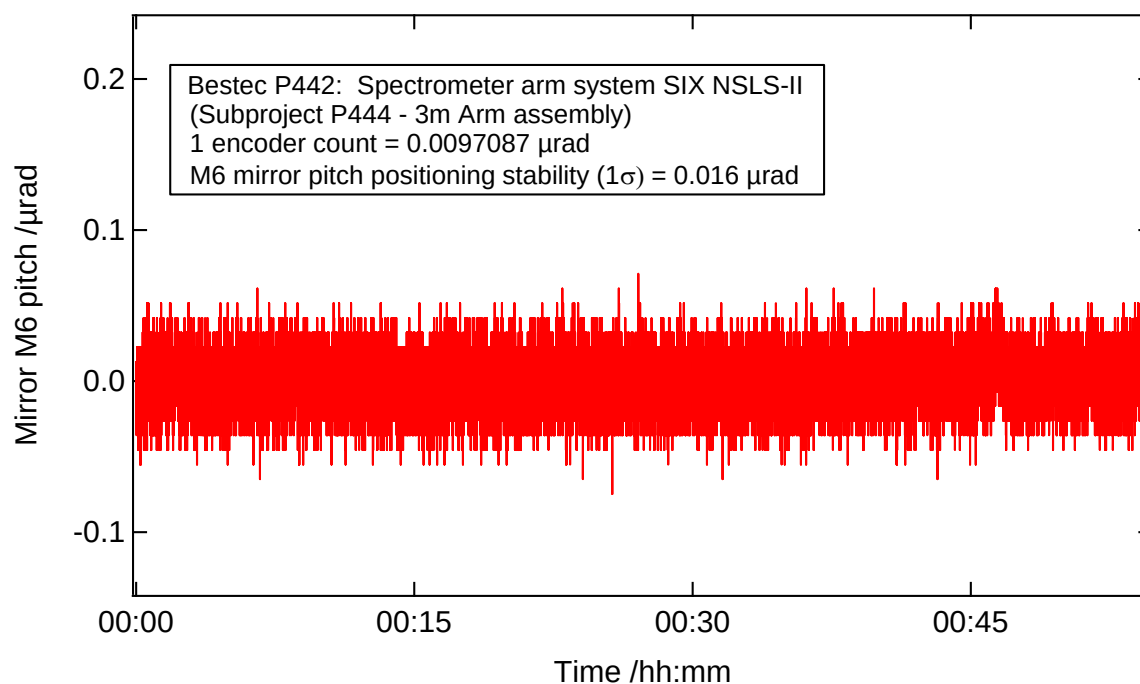


Figure 43: Long term M6 mirror pitch stability. Encoder data are neither averaged nor filtered.

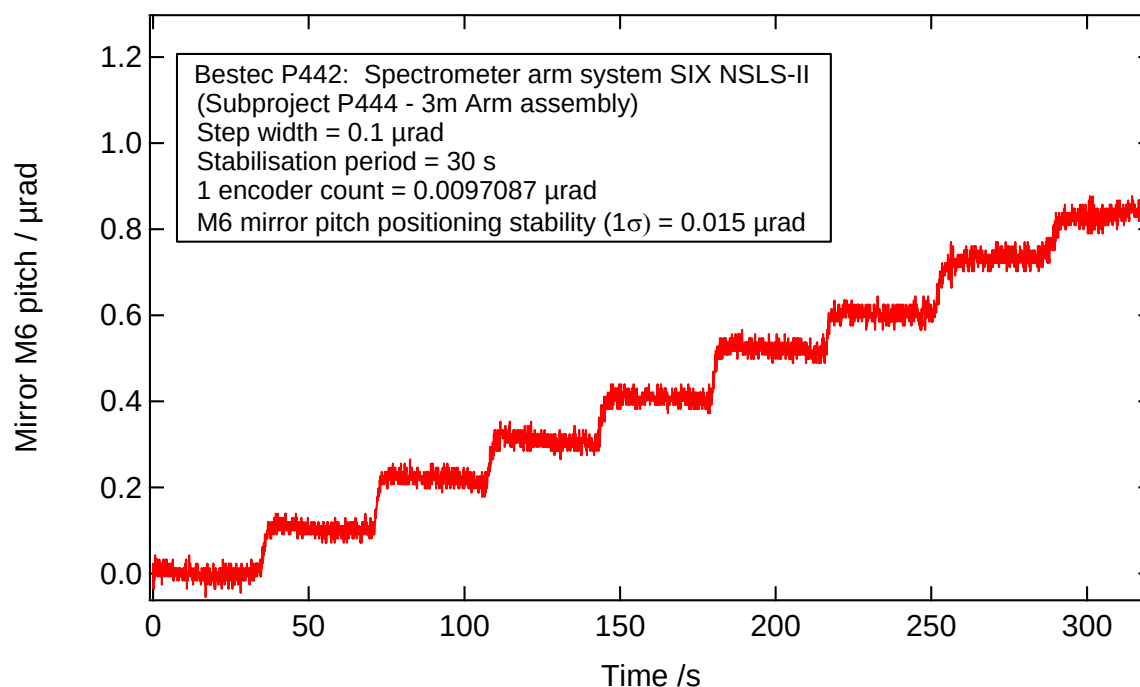


Figure 44: M6 mirror rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

Mirror M7 pitch positioning and long term stability scans

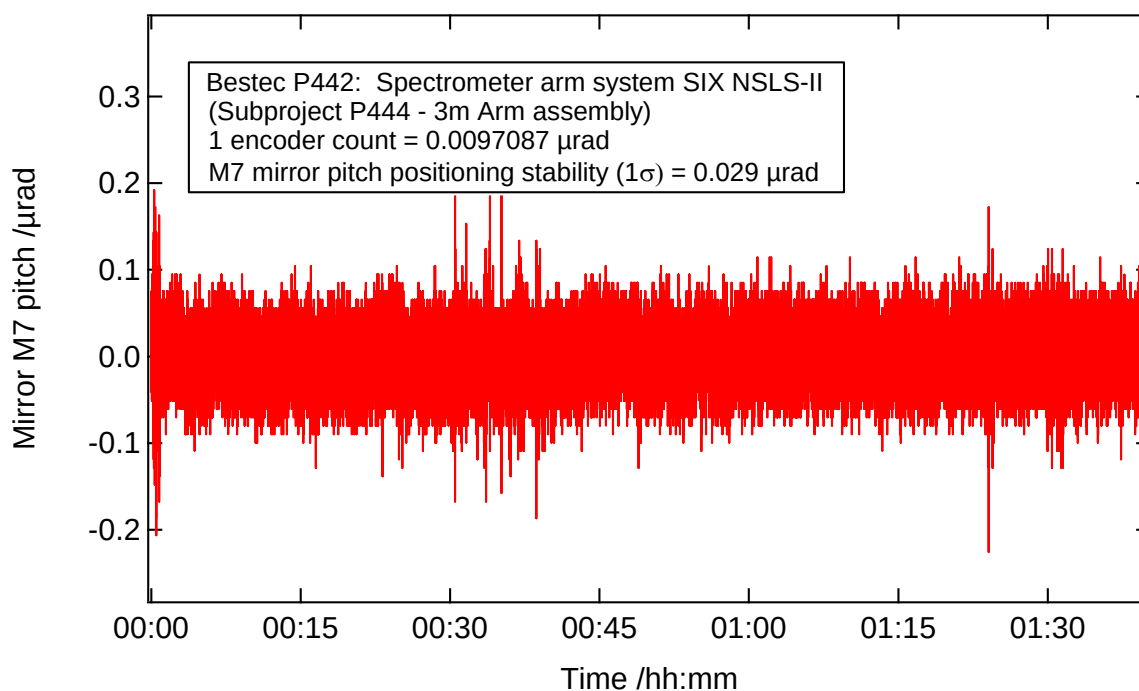


Figure 45: Long term M7 mirror pitch stability. Encoder data are neither averaged nor filtered.

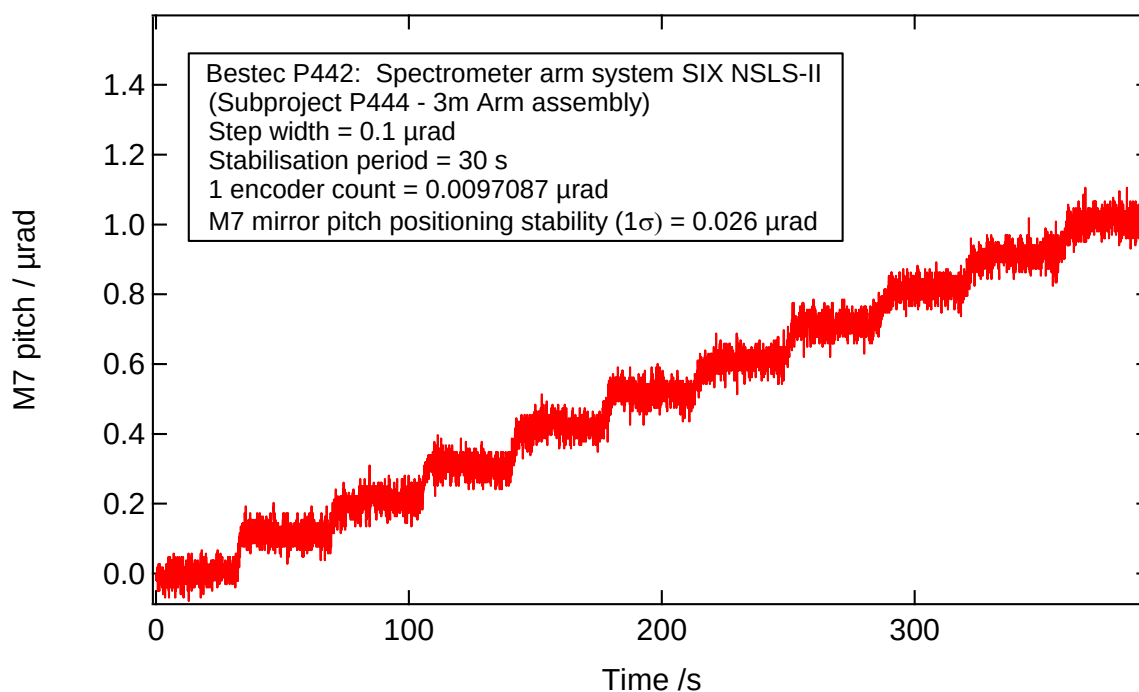


Figure 46: M7 mirror rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

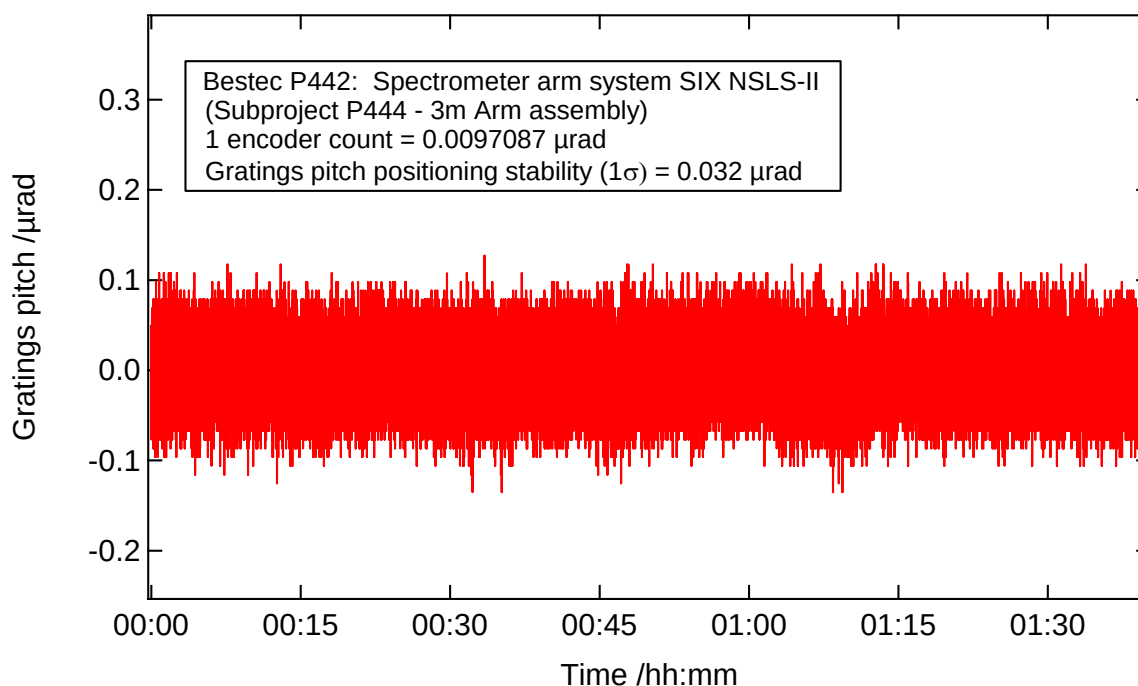
Grating pitch positioning and long term stability scans

Figure 47: Long term gratings pitch stability. Encoder data are neither averaged nor filtered.

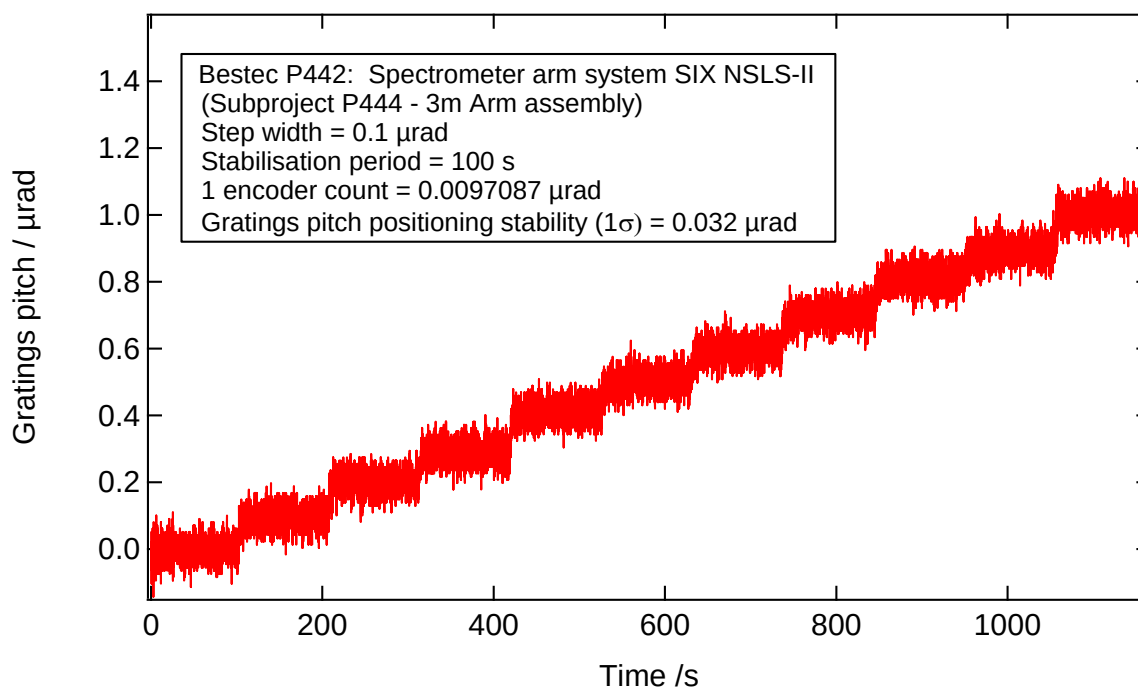


Figure 48: Gratings rotation angle (pitch) scans optimized for high resolution and stability for 0.1 μrad step width (stabilization procedure implemented in the Bestec motion control system). Encoder data are neither averaged nor filtered.

3AA 2Theta circle

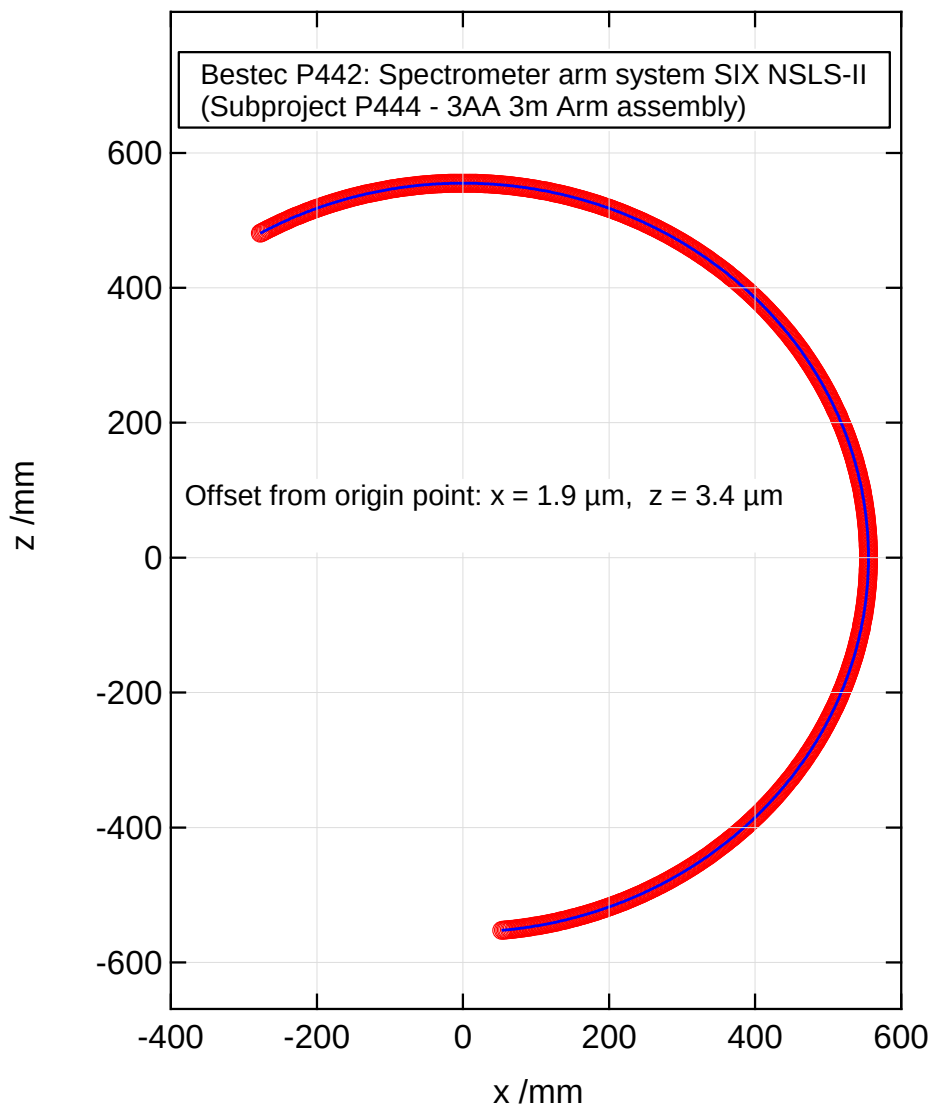


Figure 49: 3AA 2Theta circle.

The 3AA 2Theta and the M5 2Theta circles are coaxial to each other with an alignment error of $\Delta r = 81 \mu\text{m}$, which is limited by the laser tracker accuracy of $\sim 50 \mu\text{m}$.

The deviation of the average 2Theta axis of the spectrometer system from the point of origin of the global coordinate system of the spectrometer is $\sim 40 \mu\text{m}$ (also limited by the laser tracker accuracy of $\sim 50 \mu\text{m}$).

Circle of minimum confusion of the 3AA 2Theta radial bearing

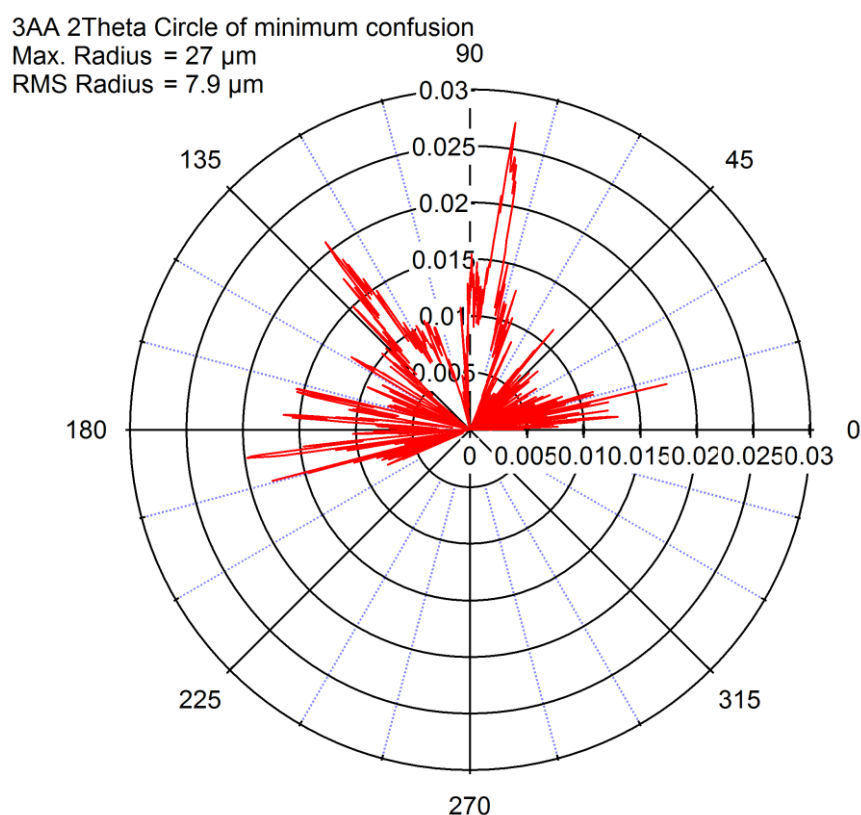


Figure 50: Circle of minimum confusion of the 3AA 2Theta radial bearing.

Height deviation along 2Theta circle of the 3AA 2Theta radial bearing

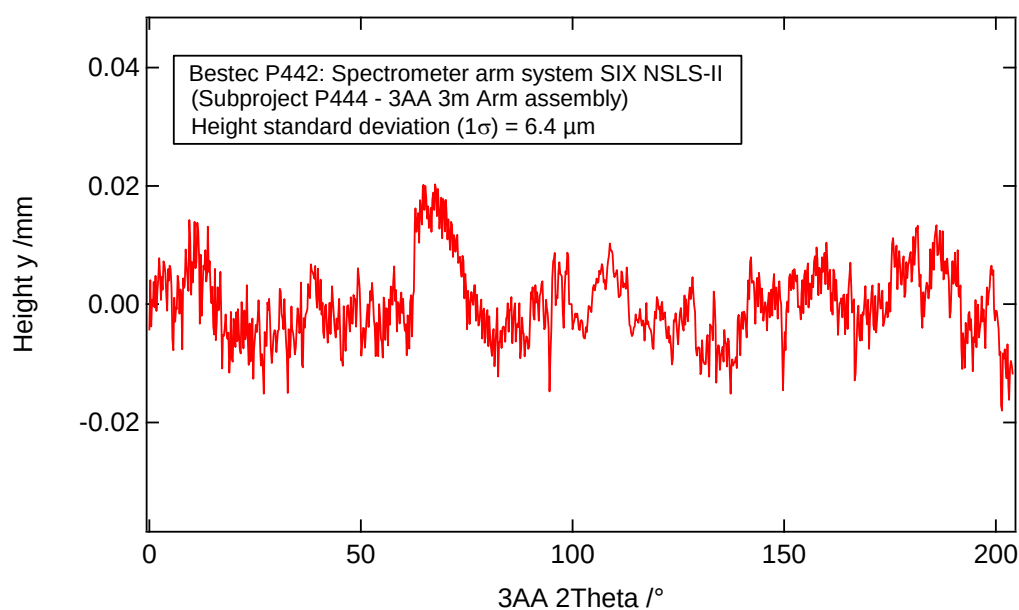


Figure 51: Height deviation along 3AA 2Theta circle of the 3AA 2Theta radial bearing.

Circle of minimum confusion of the 3AA 2Theta - coupled motion of the 3AA and DAA

Figure 52 and Figure 53 show the processed data collected during first coupled and automated movement of the 3AA and DAA 2Theta.

The spike visible in Figure 52 around 40° is related to an unintended collision of a DAA tube carriage-wheel with a temporarily installed power cord for the BNL diagnostics and is omitted in the data analysis.

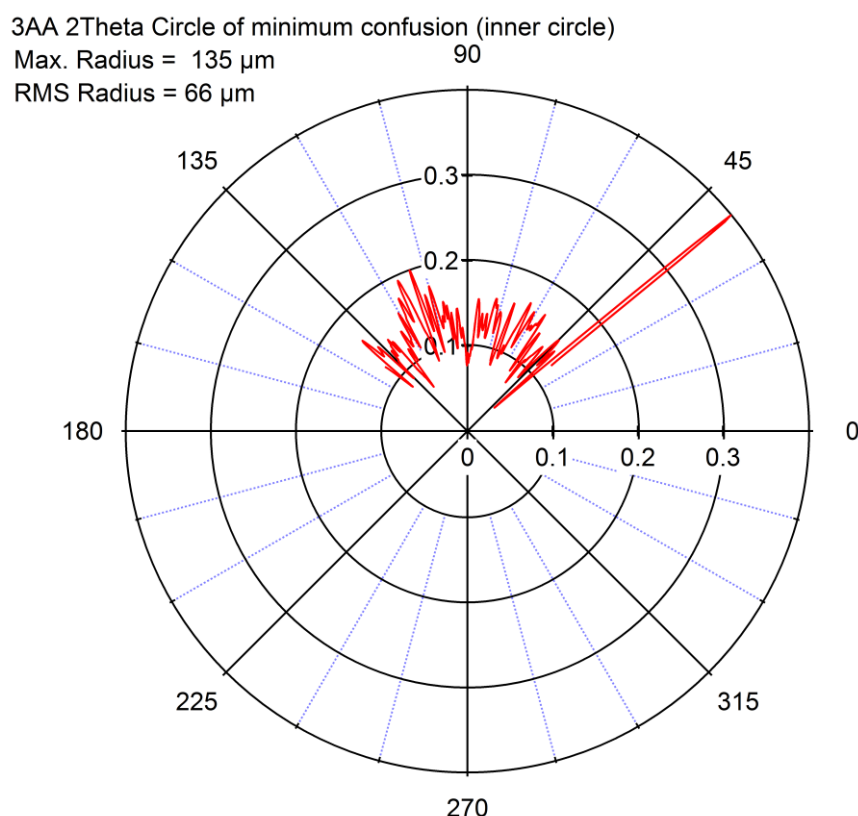


Figure 52: Circle of minimum confusion of the 3AA 2Theta. Data measured with the SMR positioned at the upstream side of the 3AA upper granite (inner circle).

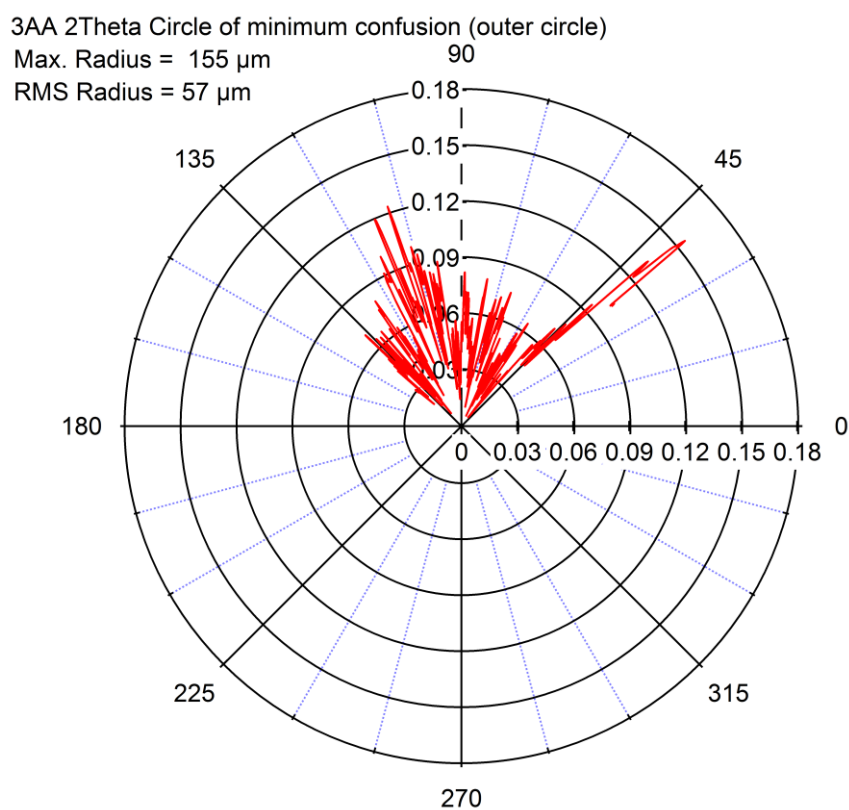


Figure 53: Circle of minimum confusion of the 3AA 2Theta. Data measured with the SMR positioned at the downstream side of the 3AA upper granite (outer circle).

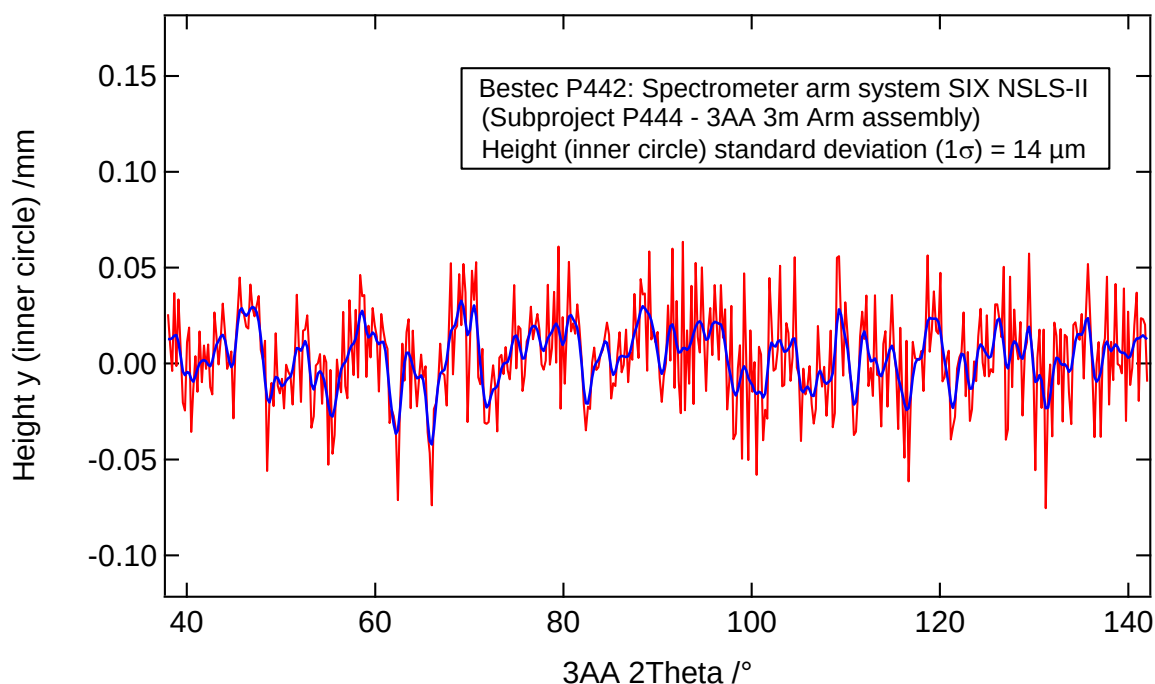
Height deviation of the 3AA 2Theta - coupled motion of the 3AA and DAA

Figure 54: Height deviation along 3AA 2Theta (raw data in red, low-pass filtered data in blue). Data measured with the SMR positioned at the upstream side of the 3AA upper granite (inner circle). The standard deviation was derived from the low-pass filtered data.

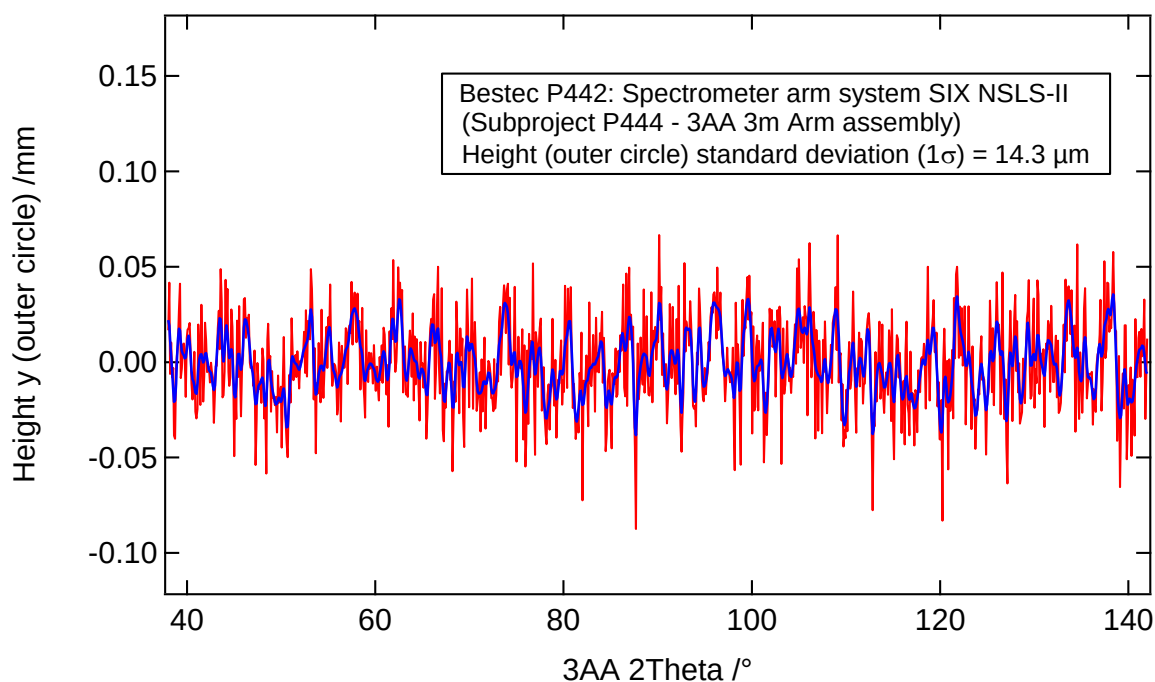


Figure 55: Height deviation along 3AA 2Theta (raw data in red, low-pass filtered data in blue). Data measured with the SMR positioned at the downstream side of the 3AA upper granite (outer circle). The standard deviation was derived from the low-pass filtered data.

3.2.3 Additional hardware tests

Tests of the electrical components

Table 10: Overview of tested electrical components

Hardware abbreviation	Test method	Status
TC7	Temperature reading	OK
TC8	Temperature reading	OK
TC9	Temperature reading	OK
TC10	Temperature reading	OK
TC11	Temperature reading	OK
TC12	Temperature reading	OK
CSW-GP	Resistance	OK
FSW-GT	Resistance	OK
CSW1-3AAR2T	Resistance	OK
CSW2-3AAR2T	Resistance	OK
CSW3-3AAR2T	Resistance	OK
CSW4-3AAR2T	Resistance	OK
FSW1-3AAR2T	Resistance	OK
FSW2-3AAR2T	Resistance	OK

3.2.4 Location of in-vacuum components and sensors

Table 11: Location of in-vacuum components and sensors

In air (feedthrough)	In vacuum	Wiring scheme
TC7	At M6 mirror holder base plate	444 - 7M - 346a
TC8	At M7 mirror cradle	444 - 7M - 346a
TC9	At the gratings translation slide	444 - 7M - 346a
TC10	At M7 / gratings support granite block	444 - 7M - 346b
TC11	At M6 support granite block	444 - 7M - 346b
TC12	At the lower granite block	444 - 7M - 346b

3.3 DAA - detector arm and stand assembly

3.3.1 Technical specifications

Table 12: Overview of technical specifications (DAA)

General	
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	cylindrical, vertical axis
Main flange (chamber lid)	COF-type, diameter 700 mm
DAA height upstream left (DAAHUL)	
Total range	51.5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height upstream right (DAAHUR)	
Total range	51.6 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height downstream (DAAHD)	
Total range	35.15 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00

	Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
2-theta rotation (DAAR2T)	
Total range	113.4°
Offset (at this position the incident beam on DAA is perpendicular to the incident beam on the sample)	15747638 counts
Motor	Phytron ZSH87/3.200.5-PLE80/20
Step resolution (full step)	3.8957×10^{-4} °/step
Encoder	Balluff Encoder scale: BML-M02-A55-A0-M3300 Readhead: BML-S1G0-B7ED-M5EZ-90-S284
Encoder resolution	3.9566798×10^{-6} ° /count
Limit switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
Collision switch	2 x Schaltbau S880
REW LSW-DAAR2T	1453122 counts
FWD LSW-DAAR2T	30111482 counts
DAA 2Theta circle Offset from origin point	x = -614 μm z = -24 μm
Circle of minimum confusion	Max. radius = 324 μm RMS radius = 114 μm
Height deviation along DAA 2Theta circle	35.5 μm
Detector translation (DT)	
Range	- 253 mm to 253 mm
Offset (middle position)	6025499 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	5 mm
Linear drive	HIWIN HGW35CC3R1220ZAPSS
Step resolution (full step)	0.5 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B10A
Encoder resolution	0.05 μm /count

Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
REW LSW-DT	959216 counts
FWD LSW-DT	11091782 counts
Aperture unit (AT, AB, AL, AR)	
Blade material	OFHC
Translation range AT	-1.9 mm to 31.9 mm
Translation range AB	-35.23 mm to 4.14 mm
Translation range AL	-1.25 mm to 46.66 mm
Translation range AR	-45.35 mm to 2.48 mm
Offset AT (at beam height 1161.585 mm)	- 30896 counts (47 mm - scale at actuator)
Offset AB (at beam height 1161.585 mm)	33203 counts (43.8 mm - scale at actuator)
Offset AL (beam center)	- 44400 counts (47.2 mm - scale at actuator)
Offset AR (beam center)	43402 counts (46.2 mm - scale at actuator)
Linear feedthrough	VAb MLS40-50
Stroke	50 mm
Motor	Phytron ZSS57.200.2,5-I with encoder
Spindle pitch	1 mm
Step resolution (full step)	10 µm /step at linear actuator
Encoder resolution	1 µm /count
Limit switches	2 x Schaltbau S880
Initialisation AT	Forward limit switch, negative search direction for reference mark
REV LSW-AT	- 32787 counts (49 mm - scale at actuator)
FWD LSW-AT	1022 counts (14.8 mm - scale at actuator)
Initialisation AB	Reverse limit switch, positive search direction for reference mark
REV LSW-AB	- 2027 counts (8.2 mm - scale at actuator)
FWD LSW-AB	37339 counts (47.8 mm - scale at actuator)
Initialisation AL	Forward limit switch, negative search direction for reference mark
REV LSW-AL	- 45745 counts (48.6 mm - scale at actuator)
FWD LSW-AL	2255 counts (0.5 mm - scale at actuator)

Initialisation AR	Reverse limit switch, positive search direction for reference mark
REV LSW-AR	-1943 counts (0.8 mm - scale at actuator)
FWD LSW-AR	45877 counts (47.8 mm - scale at actuator)

3.3.2 Test results

DAA 2Theta circle

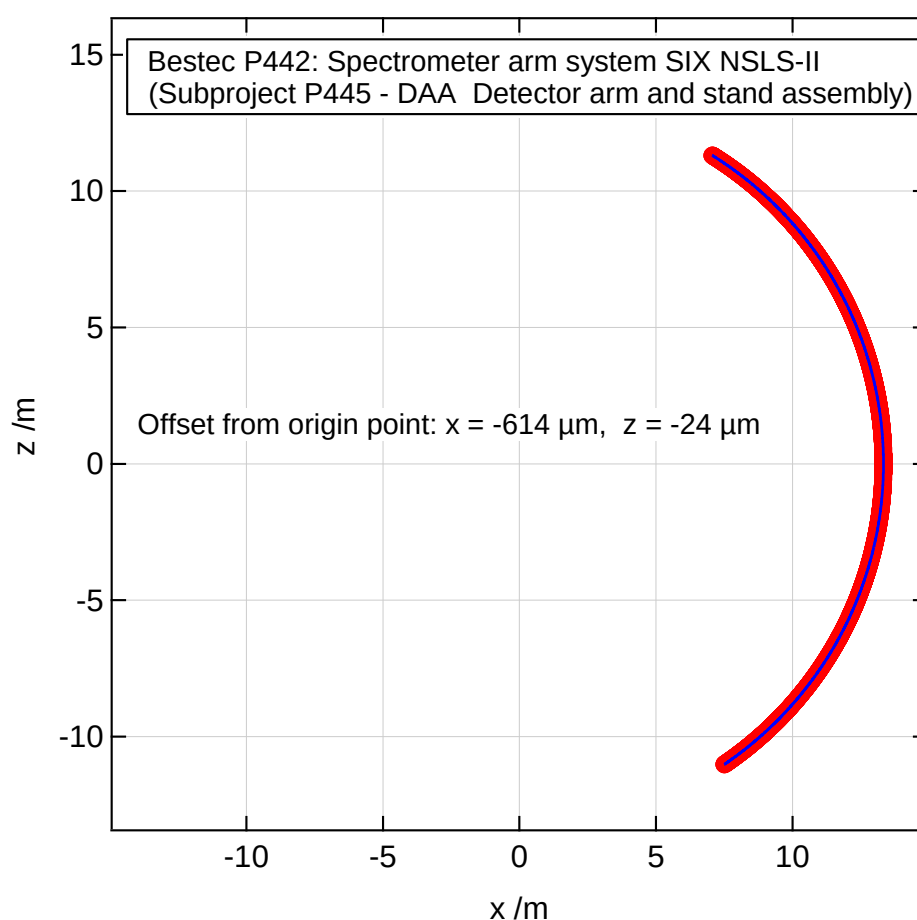


Figure 56: DAA 2Theta circle.

DAA 2Theta circle of minimum confusion

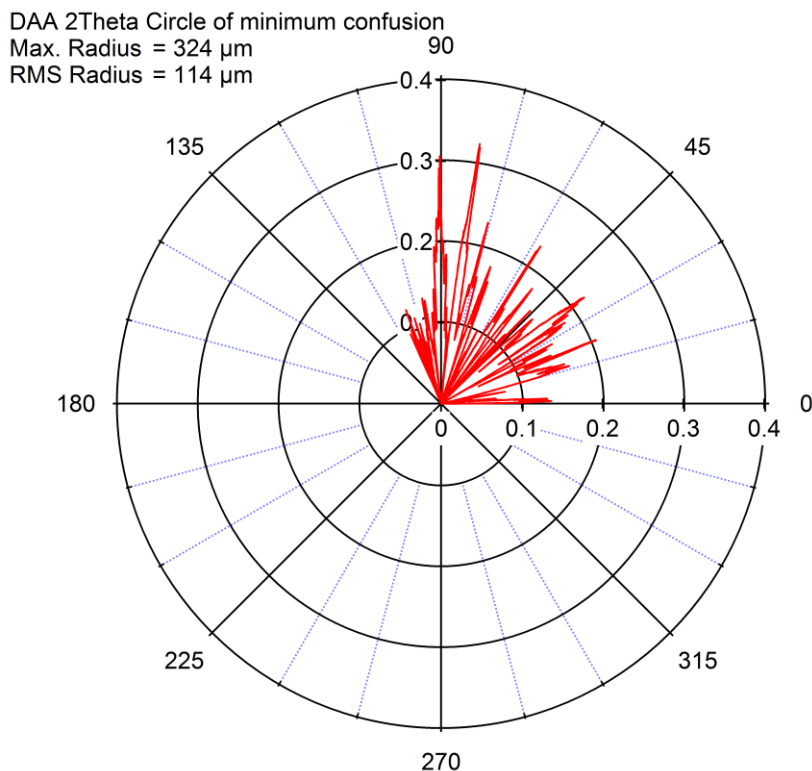


Figure 57: Circle of minimum confusion of the DAA 2Theta rotation stage.

Height deviation along DAA 2Theta circle

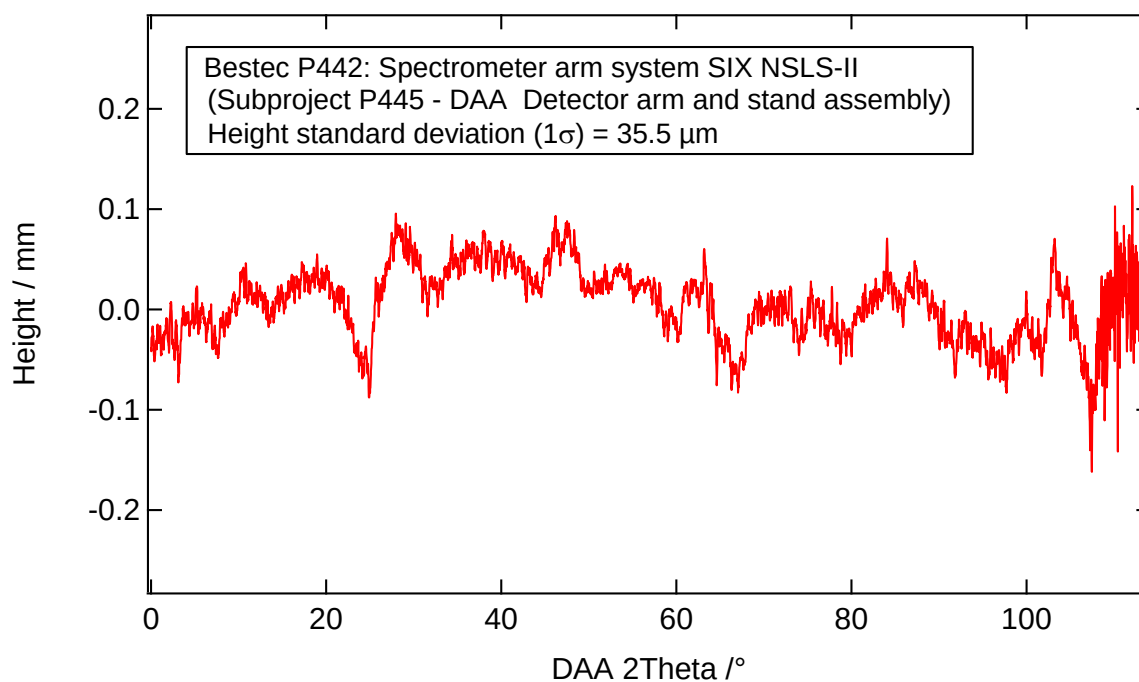


Figure 58: Height deviation along DAA 2Theta circle.

4 Conclusions

The installation and commissioning of the system 'Spectrometer arm mirror system for SIX beamline at NSLS II' according to contract 280379 between BNL and Bestec GmbH and its amendments was successfully completed.